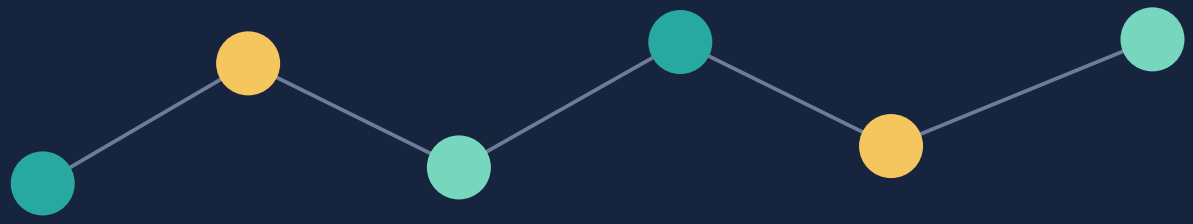
GATE 2027 / COMPLETE SYLLABUS EDITION

Computer Networks

Start from the basics. Build every syllabus idea. Then practise.

Section 10: Computer Networks



LEARN → SEE → WORK THROUGH → RETRIEVE → PRACTISE

1 Understand

Each chapter starts in plain language.

2 Visualize

Follow the flow and redraw it.

3 Work

Trace the worked example with pencil.

4 Recall

Answer before checking the solution.

A beginner learning guide + illustrated recall/practice + your supplied reference material

2027 syllabus coverage map

Use this as a route map: every item below has a beginner lesson in this PDF. The supplied handbook/chapter PDFs are gathered in the reference section after the learning and practice pages.

- [ ] **Layering, switching and performance**

  Layering • Circuit switching • Packet switching • Metrics

- [ ] **Data link: errors, MAC and Ethernet**

  Error detection • Framing • MAC • Ethernet

- [ ] **Distance-vector and link-state routing**

  Distance vector • Count to infinity • Link state • Comparison

- [ ] **IPv4, fragmentation, CIDR and NAT**

  CIDR • Subnet range • Fragmentation • NAT

- [ ] **TCP flow and congestion control**

  Sequence and ACK • Flow control • Congestion control • Connection

- [ ] **Socket API**

  Server sequence • Client sequence • Address and port • Byte stream

- [ ] **DNS and HTTP**

  DNS • HTTP request/response • Persistence and caching • HTTPS

- [ ] **Reliable delivery, windows and link efficiency**

  Stop and wait • Sliding window • Sequence number bound • TCP flow and congestion

- [ ] **IPv4 delivery: CIDR, NAT and fragmentation**

  CIDR block • Longest prefix • Fragment offset • NAT

Your route through this subject

Follow the lessons in order. Tick a topic only after you can explain its picture without looking; a tick means “I can teach it,” not “I have seen it.”

1. Layering, switching and performance
2. Data link: errors, MAC and Ethernet
3. Distance-vector and link-state routing
4. IPv4, fragmentation, CIDR and NAT
5. TCP flow and congestion control
6. Socket API
7. DNS and HTTP
8. Reliable delivery, windows and link efficiency
9. IPv4 delivery: CIDR, NAT and fragmentation

THE 20-MINUTE LEARNING LOOP

5 min	5 min	7 min	3 min
Read	Redraw	Solve	Recall

IF THIS SUBJECT FEELS COMPLETELY NEW

That is okay. Start with the first lesson, keep a pencil beside you, and redraw its picture once. Attempt the recall prompt before opening the answer guide.

Layering, switching and performance

START WITH THE IDEA

A network moves information between hosts. Layering divides the job into services; switching determines how traffic uses links and paths.

A CONCRETE FIRST EXAMPLE

A 1000-bit packet on a 1 Mbps link needs 1 ms to transmit, regardless of how far it later propagates.

PICTURE IT



IMAGINE IT THIS WAY

Layering is a delivery company: each department handles one part and hands a package to the next.

THE IDEAS, ONE BY ONE

Layering

Each layer offers a service to the layer above and uses the layer below. Encapsulation adds headers as data moves down.

Circuit switching

Reserve an end-to-end path/resources for a session.

Packet switching

Share links by sending packets; datagrams route independently, virtual circuits establish logical paths.

Metrics

Transmission delay=L/R; propagation delay=distance/speed. Also track throughput, queueing delay, processing delay, loss and end-to-end delay.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Transmission time depends on bits and rate; propagation time depends on distance and medium.

TRY FROM MEMORY

Cover this page. Explain Layering, switching and performance using the picture, then give one example before looking again.

Data link: errors, MAC and Ethernet

START WITH THE IDEA

The data-link layer delivers frames over one link. It detects corruption and coordinates access to a shared medium.

A CONCRETE FIRST EXAMPLE

CRC appends a remainder so transmitted polynomial is divisible by the generator; receiver checks whether division leaves zero remainder.

PICTURE IT



IMAGINE IT THIS WAY

A frame is a local envelope; MAC rules decide who speaks on a shared wire.

THE IDEAS, ONE BY ONE

Error detection

Parity catches some errors; checksum summarizes data; CRC treats bits as a polynomial and detects burst errors well.

Framing

Frames delimit link-layer units and carry addressing/control fields.

MAC

ALOHA/CSMA strategies manage shared-medium access; collision behavior depends on the protocol.

Ethernet

Ethernet frames use MAC addresses; classic shared Ethernet used CSMA/CD, while switched full-duplex Ethernet avoids collisions.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Error detection can flag corruption but does not automatically correct it.

TRY FROM MEMORY

Cover this page. Explain Data link: errors, MAC and Ethernet using the picture, then give one example before looking again.

Distance-vector and link-state routing

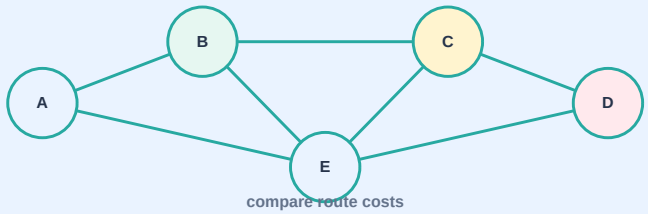
START WITH THE IDEA

Routers choose next hops. Distance-vector methods exchange destination costs with neighbors; link-state methods distribute a network map and compute routes locally.

A CONCRETE FIRST EXAMPLE

If link A -> B costs 2 and B advertises distance 5 to D, A's candidate via B is 7.

PICTURE IT



IMAGINE IT THIS WAY

Distance vector asks neighbors for directions; link state builds the whole map.

THE IDEAS, ONE BY ONE

Distance vector

Bellman-Ford style update: cost to destination via neighbor = link cost + neighbor's advertised distance.

Count to infinity

Bad news may spread slowly; split horizon/poison reverse help prevent some loops.

Link state

Flood link costs, build a topology database, then run Dijkstra from the router.

Comparison

Distance vector shares summarized routes; link state shares local links and recomputes a global view.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Do not mix routing algorithm with the transport protocol used to carry updates.

TRY FROM MEMORY

Cover this page. Explain Distance-vector and link-state routing using the picture, then give one example before looking again.

IPv4, fragmentation, CIDR and NAT

START WITH THE IDEA

IPv4 identifies interfaces with 32-bit addresses. Routers forward datagrams; subnet prefixes determine which addresses share a route.

A CONCRETE FIRST EXAMPLE

A /26 leaves 6 host bits: 64 addresses per block. Network boundaries increment in the last relevant octet by 64.

PICTURE IT



IMAGINE IT THIS WAY

CIDR is a cut in the address bits; NAT is a receptionist translating inside names to an outside identity.

THE IDEAS, ONE BY ONE

CIDR

/p means p network bits and 32-p host bits. Block size is 2^(32-p) addresses.

Subnet range

Network address has host bits 0; directed broadcast has host bits 1, subject to special cases.

Fragmentation

A datagram may be split to fit link MTU. Fragment offsets use 8-byte units; fragments are reassembled at destination.

NAT

Network address translation rewrites addresses and often ports so private hosts share public addresses.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Fragment offset is measured in 8-byte units, not raw bytes.

TRY FROM MEMORY

Cover this page. Explain IPv4, fragmentation, CIDR and NAT using the picture, then give one example before looking again.

TCP flow and congestion control

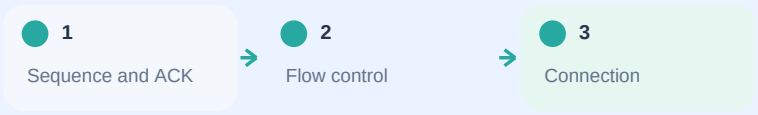
START WITH THE IDEA

TCP provides a reliable ordered byte stream. Flow control protects the receiver; congestion control protects the network.

A CONCRETE FIRST EXAMPLE

If receiver window is 5000 bytes and congestion window is 3000, sender's usable window is at most min(5000,3000)=3000 bytes.

PICTURE IT



IMAGINE IT THIS WAY

Flow control is the customer's receiving capacity; congestion control is traffic capacity on the road.

THE IDEAS, ONE BY ONE

Sequence and ACK

Sequence numbers label bytes; cumulative ACKs indicate the next byte expected.

Flow control

Receiver-advertised window limits data in flight so its buffer is not overwhelmed.

Congestion control

Sender adjusts congestion window based on perceived network capacity; slow start and congestion avoidance grow it differently.

Connection

Three-way handshake synchronizes initial sequence numbers; FIN/RST/timeout handle closure/failure.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Receiver window and congestion window solve different problems.

TRY FROM MEMORY

Cover this page. Explain TCP flow and congestion control using the picture, then give one example before looking again.

Socket API

START WITH THE IDEA

A socket is an endpoint used by an application to send or receive data over a transport protocol.

A CONCRETE FIRST EXAMPLE

A server listening on a port accepts a connection; the accepted socket represents that particular client session.

PICTURE IT



IMAGINE IT THIS WAY

IP is the building address; port is the apartment; socket is the live communication endpoint.

THE IDEAS, ONE BY ONE

Server sequence

Create socket, bind address, listen, accept a connection, read/write, close.

Client sequence

Create socket, connect to server, read/write, close.

Address and port

IP identifies a host/interface; a port identifies an application endpoint on that host.

Byte stream

TCP socket reads may return fewer bytes than requested; programs loop and frame application messages.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT TCP preserves byte order, not application message boundaries.

TRY FROM MEMORY

Cover this page. Explain Socket API using the picture, then give one example before looking again.

DNS and HTTP

START WITH THE IDEA

Application protocols define the messages applications exchange. DNS maps names to records; HTTP transfers web resources.

A CONCRETE FIRST EXAMPLE

A browser may resolve example.com to an address via DNS, then send an HTTP GET to the server's port 80 (HTTPS typically 443).

PICTURE IT



IMAGINE IT THIS WAY

DNS is a distributed contact list; HTTP is the request form exchanged with the contact.

THE IDEAS, ONE BY ONE

DNS

A distributed hierarchy resolves names through caches and authoritative servers; records include A/AAAA, CNAME, MX and others.

HTTP request/response

Requests contain method, target, headers and optional body; responses include status, headers and content.

Persistence and caching

Persistent connections reuse transport sessions; cache directives can avoid repeated transfers.

HTTPS

HTTP over TLS adds authentication, confidentiality and integrity for the connection.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT DNS name resolution and HTTP resource retrieval are separate application-layer steps.

TRY FROM MEMORY

Cover this page. Explain DNS and HTTP using the picture, then give one example before looking again.

Reliable delivery, windows and link efficiency

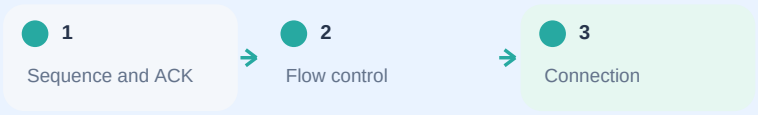
START WITH THE IDEA

A reliable link protocol numbers frames and acknowledgements so it can detect loss or duplication. Larger windows keep the link busy during propagation delay.

A CONCRETE FIRST EXAMPLE

If frame transmission time is 1 ms and round-trip time is 10 ms, stop-and-wait sends at most one frame per 11 ms cycle before considering ACK transmission.

PICTURE IT



IMAGINE IT THIS WAY

A sliding window is a train of frames already allowed onto the track while earlier acknowledgements return.

THE IDEAS, ONE BY ONE

Stop and wait

One frame may be in flight. Efficiency is low when round-trip propagation is large compared with frame transmission time.

Sliding window

A sender may keep W unacknowledged frames in flight; throughput is bounded by link rate and the bandwidth-delay product.

Sequence number bound

With k sequence bits there are 2^k sequence values. GBN sender window is at most 2^k-1; Selective Repeat needs window no larger than 2^(k-1).

TCP flow and congestion

Receiver window limits what the receiver can buffer; congestion window limits what the network can safely carry. The sender uses the smaller limit.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Do not use one sequence-number formula for both Go-Back-N and Selective Repeat.

TRY FROM MEMORY

Cover this page. Explain Reliable delivery, windows and link efficiency using the picture, then give one example before looking again.

IPv4 delivery: CIDR, NAT and fragmentation

START WITH THE IDEA

Routers forward by longest-prefix match. IPv4 may fragment a datagram to fit a link MTU; offsets locate fragments within the original payload.

A CONCRETE FIRST EXAMPLE

For MTU 1500 and a 20-byte IPv4 header, the payload limit is 1480 bytes, already divisible by 8; split payload into chunks no larger than this.

PICTURE IT



IMAGINE IT THIS WAY

CIDR is a set of addresses sharing a prefix; fragmentation is cutting a parcel into link-sized pieces with byte offsets.

THE IDEAS, ONE BY ONE

CIDR block

A /p prefix fixes p high-order bits. A block contains 2^(32-p) addresses.

Longest prefix

When several routing entries match, choose the one with the greatest prefix length.

Fragment offset

IPv4 fragment offsets are in 8-byte units. All fragments except the final one carry payload lengths divisible by 8.

NAT

NAT rewrites address/port fields and updates the IPv4 header checksum when the header changes.

HOW TO USE IT ON A PROBLEM

Label the layer, addresses, packet size and link rate. Draw the route/flow, calculate delay or window components separately, and check units and protocol assumptions.

WATCH OUT Compute fragment payload capacity after subtracting headers, then round down to a multiple of 8.

TRY FROM MEMORY

Cover this page. Explain IPv4 delivery: CIDR, NAT and fragmentation using the picture, then give one example before looking again.

Past-paper trail and coverage check

WHAT THE PAPERS MATCH

Verified anchors: 2018 Q13-Q14/Q25/Q54-Q55 (protocol fields, TCP congestion, sequence numbers, fragmentation and MAC); 2022 Q22/Q35/Q55/Q57/Q60 (Ethernet, DNS, routing and sequence numbers); 2023 Q16/Q24/Q49/Q51/Q64; 2024 CS1 Q16/Q29/Q31/Q36/Q58/Q65; 2025 CS1 Q16/Q22/Q40/Q57 and CS2 Q16-Q18/Q23/Q36; 2026 CS1 Q18-Q19/Q44-Q45/Q56 and CS2 Q21-Q22/Q33/Q43/Q58/Q65.

WHAT THIS EDITION ADDS

Added bandwidth-delay window intuition, sequence-number bounds, longest-prefix routing, NAT, IPv4 fragmentation and TCP flow-vs-congestion windows.

SYLLABUS BOUNDARY

The supplied 2027 syllabus explicitly lists IPv4, distance-vector/link-state routing, TCP, DNS and HTTP; IPv6 and BGP are optional extras, not core coverage.

HOW TO USE THIS CHECK

A question is a match when its tested skill is taught in the lesson pages, even if the story or numbers are new. Anchors are selected verified examples, not an exhaustive list; the separate Coverage and Gap Audit PDF identifies years whose scans could not be checked question by question.

Close the notes: try before you peek

For each card, explain the idea from memory and recreate the lesson example on paper.

1. **Layering, switching and performance**

   Without notes: define Metrics and explain why this example works. Then recreate the steps on the lesson page.

2. **Data link: errors, MAC and Ethernet**

   Without notes: define Ethernet and explain why this example works. Then recreate the steps on the lesson page.

3. **Distance-vector and link-state routing**

   Without notes: define Comparison and explain why this example works. Then recreate the steps on the lesson page.

4. **IPv4, fragmentation, CIDR and NAT**

   Without notes: define NAT and explain why this example works. Then recreate the steps on the lesson page.

5. **TCP flow and congestion control**

   Without notes: define Connection and explain why this example works. Then recreate the steps on the lesson page.

6. **Socket API**

   Without notes: define Byte stream and explain why this example works. Then recreate the steps on the lesson page.

Close the notes: try before you peek

For each card, explain the idea from memory and recreate the lesson example on paper.

7 **DNS and HTTP**

Without notes: define HTTPS and explain why this example works. Then recreate the steps on the lesson page.

8 **Reliable delivery, windows and link efficiency**

Without notes: define TCP flow and congestion and explain why this example works. Then recreate the steps on the lesson page.

9 **IPv4 delivery: CIDR, NAT and fragmentation**

Without notes: define NAT and explain why this example works. Then recreate the steps on the lesson page.

Compare with the worked examples

Use this only after attempting the recall prompts. The full reasoning is on the numbered lesson page.

1. **LESSON 1: Layering, switching and performance**

   A 1000-bit packet on a 1 Mbps link needs 1 ms to transmit, regardless of how far it later propagates.

2. **LESSON 2: Data link: errors, MAC and Ethernet**

   CRC appends a remainder so transmitted polynomial is divisible by the generator; receiver checks whether division leaves zero remainder.

3. **LESSON 3: Distance-vector and link-state routing**

   If link A -> B costs 2 and B advertises distance 5 to D, A's candidate via B is 7.

4. **LESSON 4: IPv4, fragmentation, CIDR and NAT**

   A /26 leaves 6 host bits: 64 addresses per block. Network boundaries increment in the last relevant octet by 64.

5. **LESSON 5: TCP flow and congestion control**

   If receiver window is 5000 bytes and congestion window is 3000, sender's usable window is at most min(5000,3000)=3000 bytes.

6. **LESSON 6: Socket API**

   A server listening on a port accepts a connection; the accepted socket represents that particular client session.

Compare with the worked examples

Use this only after attempting the recall prompts. The full reasoning is on the numbered lesson page.

7. LESSON 7: DNS and HTTP

A browser may resolve example.com to an address via DNS, then send an HTTP GET to the server’s port 80 (HTTPS typically 443).

8. LESSON 8: Reliable delivery, windows and link efficiency

If frame transmission time is 1 ms and round-trip time is 10 ms, stop-and-wait sends at most one frame per 11 ms cycle before considering ACK transmission.

9. LESSON 9: IPv4 delivery: CIDR, NAT and fragmentation

For MTU 1500 and a 20-byte IPv4 header, the payload limit is 1480 bytes, already divisible by 8; split payload into chunks no larger than this.

Your practice and reference library

01 **Beginner lessons**

Plain-language explanation, visual sequence, example and watch-outs for every listed 2027 topic cluster.

02 **Recall and GATE-style practice**

The illustrated recap section includes questions and a separate answer explanation.

03 **Detailed reference**

Your supplied subject handbook and chapter PDFs follow for extra depth, reading and worked material.

Included source documents

Illustrated recall and GATE-style practice	CN - Computer Networks - GATE 2027 Illustrated Notes.pdf	12 pages
Detailed supplied reference	CN.pdf	40 pages
Detailed supplied reference	Computer-Networks-GATE-2027-Master-Study-Sheet.pdf	11 pages

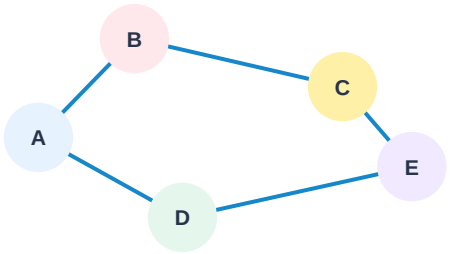
The official 2027 syllabus map precedes the lessons. Original reference pages are retained so you can use them for slower reading and extra practice.

Computer Networks

GATE 2027 | VISUAL LEARNING NOTES | STARTING FROM THE BASICS

Follow a message across layers, links, routes and endpoints.

PICTURE IT



each layer adds one service

HOW TO USE THIS NOTEBOOK

1. **Learn the idea**
   Read the plain-language explanation before memorizing a formula.
2. **See it**
   Study the labeled picture and redraw it from memory.
3. **Follow a solve**
   Work through the examples step by step.
4. **Test yourself**
   Attempt the GATE-style questions before opening the key.

THE PATH YOU WILL FOLLOW

- Architecture and switching
- Link and routing
- Internet, transport and application

IF YOU FEEL COMPLETELY NEW

That is okay. Start with the first topic page, keep a pencil beside you, and copy each diagram once. Do not jump straight to the answer key. The questions are meant to come after the concepts.

Your route through this subject

Follow the topics in order. Each page teaches one idea, shows a picture, gives a recall cue and points out a common trap.

ARCHITECTURE AND SWITCHING

- [ ] Principles of layering.
- [ ] Circuit, packet and virtual-circuit switching.
- [ ] Performance metrics.

LINK AND ROUTING

- [ ] Error detection.
- [ ] Medium Access Control and Ethernet.
- [ ] Distance-vector and link-state routing.

INTERNET, TRANSPORT AND APPLICATION

- [ ] IPv4 fragmentation, CIDR notation and Network Address Translation.
- [ ] TCP flow control, congestion control and socket API.
- [ ] DNS and HTTP.

THE 20-MINUTE LEARNING LOOP

5 min	5 min	7 min	3 min
Read	Redraw	Solve	Recall

Tick a topic only after you can explain its picture without looking. A tick means "I can teach it," not "I have seen it."

BEFORE EACH STUDY SESSION

Pick one topic. Read its idea, redraw its diagram, follow the worked examples, then answer questions with the notes closed. Circle only the step where you got stuck - that is your next revision target.

CONCEPT 01 / CN

Layering and switching

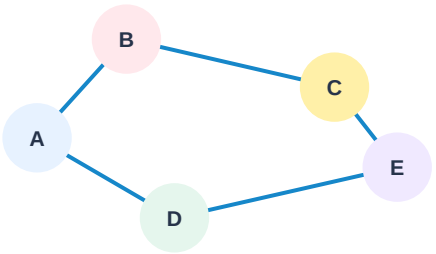
START WITH THE IDEA

Layers provide services through interfaces; encapsulation adds headers as data moves down. Circuit switching reserves resources; packet switching shares links per packet; virtual circuits set up a logical route.

A CONCRETE FIRST EXAMPLE

Layers split communication into jobs with clear interfaces. As data moves down the stack, each layer adds control information; the receiver removes those headers in reverse order. An HTTP message becomes a TCP segment, then an IP packet, then a link-layer frame; each layer adds its own header.

PICTURE IT



each layer adds one service

IMAGINE IT THIS WAY

Parcel delivery: application creates the message; lower layers add local delivery labels.

HOW TO USE IT ON A PROBLEM

For a path question, identify which layer owns each address and function.

WATCH OUT

Do not confuse a service/interface with a protocol implementation.

TRY FROM MEMORY

Cover the page and explain Layering and switching using the diagram. Give one example before looking again.

CONCEPT 02 / CN

Performance metrics

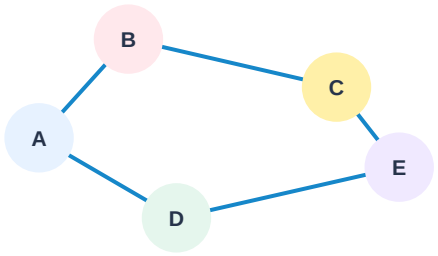
START WITH THE IDEA

Transmission delay=L/R; propagation delay=distance/propagation speed. Queueing and processing add delay. Throughput is achieved rate; bandwidth is capacity; utilization is busy fraction.

A CONCRETE FIRST EXAMPLE

Delay has several parts: transmission time depends on packet bits and link rate; propagation time depends on distance and signal speed. Queueing and processing add more delay. A 12000-bit packet on a 2 Mbps link takes 12000/2000000=0.006 s, or 6 ms, just to transmit.

PICTURE IT



transmit + propagate + queue

IMAGINE IT THIS WAY

A road has a speed limit (capacity) and traffic delay (queueing); a long road adds propagation time.

HOW TO USE IT ON A PROBLEM

Keep bits, bits/s, metres and seconds consistent.

WATCH OUT

High bandwidth does not imply low propagation delay.

TRY FROM MEMORY

Cover the page and explain Performance metrics using the diagram. Give one example before looking again.

CONCEPT 03 / CN

Error detection, MAC and Ethernet

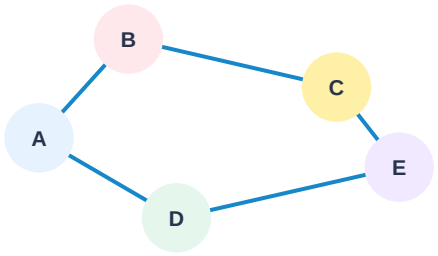
START WITH THE IDEA

Parity, checksum and CRC detect classes of errors; CRC treats bits as a polynomial over GF(2). MAC coordinates shared links. Ethernet uses frame fields and addresses; collision behavior depends on half/full duplex and technology.

A CONCRETE FIRST EXAMPLE

Parity, checksums and CRCs detect errors but do not normally correct them. CRC performs polynomial division over bits using XOR; MAC protocols decide who may use a shared link. For binary long division, subtract means XOR: 1011 XOR 1101 = 0110. The final remainder is appended to the frame.

PICTURE IT



divide bits with XOR

IMAGINE IT THIS WAY

A CRC is a tamper-evident seal: it detects many changes but does not repair them.

HOW TO USE IT ON A PROBLEM

For CRC, append zeros, divide by generator with XOR, append remainder.

WATCH OUT

Detection is not correction; modern switched full-duplex Ethernet has no CSMA/CD collision process.

TRY FROM MEMORY

Cover the page and explain Error detection, MAC and Ethernet using the diagram. Give one example before looking again.

CONCEPT 04 / CN

Distance-vector and link-state routing

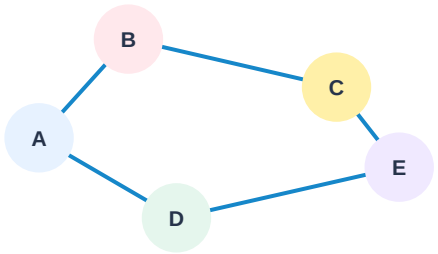
START WITH THE IDEA

Distance-vector routers exchange destination costs with neighbors (Bellman-Ford style). Link-state routers flood topology and run shortest-path computation locally. Convergence, loops and stale information matter.

A CONCRETE FIRST EXAMPLE

Distance-vector routers learn costs from neighbors; link-state routers flood a map and compute shortest paths locally. Routing is about reaching a destination; an MST is a different graph task. In distance-vector, router u updates D(u,v)=min over neighbors w of [cost(u,w)+D(w,v)].

PICTURE IT



routers exchange route knowledge

IMAGINE IT THIS WAY

Distance-vector asks neighbors for directions; link-state builds the whole road map.

HOW TO USE IT ON A PROBLEM

State the update rule, metric and convergence assumption.

WATCH OUT

Do not equate routing protocol names with algorithms unless the question says so.

TRY FROM MEMORY

Cover the page and explain Distance-vector and link-state routing using the diagram. Give one example before looking again.

CONCEPT 05 / CN

IPv4 fragmentation, CIDR and NAT

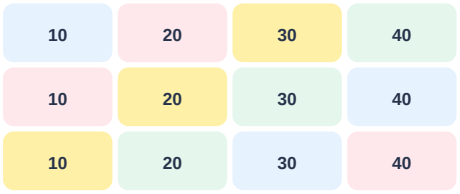
START WITH THE IDEA

CIDR /p fixes p network bits; address block size is 2^(32-p). IPv4 fragmentation respects MTU and 8-byte offset units. NAT rewrites address/port mappings between private and external networks.

A CONCRETE FIRST EXAMPLE

CIDR /p reserves p prefix bits for the network, leaving 32-p host bits. Fragment offsets use 8-byte units. NAT rewrites private address/port pairs to share public connectivity. A /26 block has 2■=64 addresses. For traditional IPv4 subnetting, subtract network and broadcast for 62 usable host addresses.

PICTURE IT



prefix fixes network bits

IMAGINE IT THIS WAY

CIDR draws a fence around address blocks; NAT is a building receptionist mapping room numbers to public lines.

HOW TO USE IT ON A PROBLEM

Hosts per subnet: block size minus network/broadcast where traditional subnet rules apply.

WATCH OUT

Fragment offset is in 8-byte units; do not subtract a header twice.

TRY FROM MEMORY

Cover the page and explain IPv4 fragmentation, CIDR and NAT using the diagram. Give one example before looking again.

CONCEPT 06 / CN

TCP flow control, congestion control and socket API

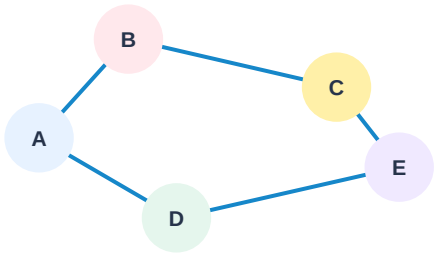
START WITH THE IDEA

Flow control protects the receiver using advertised window; congestion control protects the network using congestion window. Sequence/ACK numbers track bytes. Socket calls create/bind/listen/accept or connect endpoints.

A CONCRETE FIRST EXAMPLE

TCP numbers bytes, acknowledges received data and uses windows. Receiver flow control limits what the receiver can accept; congestion control limits what the network can safely carry. A server calls socket, bind, listen, accept. A client calls socket, then connect; send/receive follow after connection.

PICTURE IT



sequence + acknowledgements

IMAGINE IT THIS WAY

Receiver window is the receiver's inbox; congestion window is the network's safe road capacity.

HOW TO USE IT ON A PROBLEM

Use min(rwnd,cwnd) as the send-window limit in the usual model.

WATCH OUT

Flow control and congestion control solve different bottlenecks.

TRY FROM MEMORY

Cover the page and explain TCP flow control, congestion control and socket API using the diagram. Give one example before looking again.

CONCEPT 07 / CN

DNS and HTTP

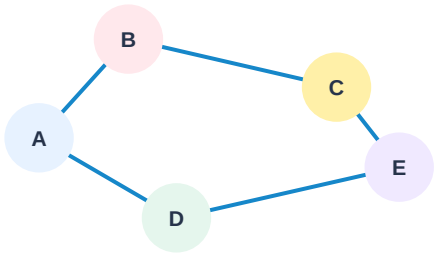
START WITH THE IDEA

DNS maps names to records through a hierarchy and caching. HTTP is an application protocol with methods, status codes, headers and representations; transport choice depends on version/context.

A CONCRETE FIRST EXAMPLE

DNS looks up names through a hierarchy and caches answers. HTTP defines requests and responses; name lookup, transport connection and the HTTP exchange are separate steps. A browser asks DNS for a server address, connects to it, then sends an HTTP request such as GET /index.html.

PICTURE IT



name -> record -> address

IMAGINE IT THIS WAY

DNS is the contact list; HTTP is the conversation format once the contact is found.

HOW TO USE IT ON A PROBLEM

Follow name lookup, connection and request/response as separate steps.

WATCH OUT

DNS resolution and HTTP request are distinct transactions.

TRY FROM MEMORY

Cover the page and explain DNS and HTTP using the diagram. Give one example before looking again.

Let's solve it together

Cover the answer with your hand. Try each step first, then compare your reasoning.

EXAMPLE 1: CIDR block

QUESTION

For 192.0.2.64/26, how many total addresses are in the block?

WORK THROUGH THE STEPS

Host bits=32-26=6. Block size=2^6=64. The block

ANSWER 64 total addresses; under conventional subnet use, 62 host addresses.

Check: write the key rule beside your answer. If it does not fit the rule, redo the model before recalculating.

EXAMPLE 2: Delay components

QUESTION

A 12000-bit packet crosses a 2 Mbps link. Find transmission delay. Ignore other delays.

WORK THROUGH THE STEPS

Transmission delay=L/R=12000/2,000,000

ANSWER 6 ms.

Check: write the key rule beside your answer. If it does not fit the rule, redo the model before recalculating.

Now you try

Attempt these without the notes. The question types are marked MCQ, MSQ or NAT. Answers and explanations are on the next page.

NAT A /26 IPv4 block contains how many total addresses?

Enter an integer.

MCQ Which control limits sending based on receiver buffer availability?

(A) Congestion window (B) Receiver advertised window (C) TTL (D) DNS cache

MCQ A TDM/circuit-switched call typically:

(A) Reserves a resource path (B) Sends each packet independently (C) Has no setup (D) Uses CRC to route

MSQ Which are true?

(A) Link-state routers distribute topology information (B) Distance-vector uses neighbor information (C) CRC corrects every error (D) TCP congestion control protects the network

MCQ Which is the normal TCP server sequence for a passive listening socket?

(A) accept, listen, bind, socket (B) socket, bind, listen, accept (C) connect, accept, socket, bind (D) DNS, HTTP, bind, listen

Exam habit: decide what is being asked before doing arithmetic. Show units, assumptions and one line of reasoning.

Answers with the why

Read the explanation even when your answer is correct. The goal is to learn the reasoning pattern.

1. 64

WHY

There are 6 host bits, so 2^6=64 addresses.

2. B

WHY

The advertised receive window implements TCP flow control.

3. A

WHY

Circuit switching reserves resources for the connection.

4. A, B, D

WHY

CRC detects errors; it does not correct every error.

5. B

WHY

Create the socket, bind its local address, listen, then accept connections.

Review loop: mark each missed concept page, redraw its picture, and retry a similar problem tomorrow.

Computer Networks



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Computer Networks

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1 IP ADDRESSING, SUBNETTING & SUPERNETTING

1.1 IP Addressing

Class A : 0 → (1 - 126), No. of IP Addresses = 2^{31}
Class B : 10 → (128 - 191), No. of IP Addresses = 2^{30}
Class C : 110 → (192 - 223), No. of IP Addresses = 2^{29}
Class D : 1110 → (224 - 239), No. of IP Addresses = 2^{28}
Class E : 1111 → (240 - 255), No. of IP Addresses = 2^{28}

1.2 Default subnet Mask

Class A : 255.0.0.0
Class B : 255.255.0.0
Class C : 255.255.255.0

1.3 Private Addresses Range

10.0.0.0 to 10.255.255.255 → 1 class A Network.
172.16.0.0 to 172.31.255.255 → 16 class B Network.
192.168.0.0 to 192.168.255.255 → 256 class C Network.

Class	Number of Networks	Number of hosts per Network
Class A	$2^7 - 2 = 126$	$2^{24} - 2 = 1,67,77,214$ hosts
Class B	$2^{14} = 16,384$	$2^{16} - 2 = 65,534$ hosts
Class C	$2^{21} = 20,97,125$	$2^8 - 2 = 254$ hosts
Class D	No NID and HID, all 28 remaining bits are used to define multicast address	
Class E	No NID and HID, it is meant for research and future purpose	

Note:
The IP address 127.x.y.z is known as loop back address and it is used to check the connectivity.

1.4 Types of Communication

(i) Unicast communication (1 : 1)

(ii) Broadcast communication (1 : All)

(iii) Multicast Communication (1: Many)

1.5 Unicast Communication

1. Transmitting the data from one computer to another computer is called as unicast communication.
2. It is one to one transmission.
3. In Unicast communication both source and destination either present in the same network or in the different network.

1.6 Broadcast Communication



1.6.1 Limited Broadcasting

1. Transmitting data from one computer to all other computer in the same network is called as Limited Broadcasting.
2. Limited Broadcast Address = 255.255.255.255
3. Limited broadcast address can't be used as a source IP Address.
4. Limited broadcast Address will always be used as a Destination IP.

1.6.2 Direct Broadcasting

1. Transmitting data from one computer to all other computer in the different network is called as Limited Broadcasting.
2. Direct broadcast address can't be used as a source IP Address.
3. Direct broadcast Address will always be used as a Destination IP.

	NID	**HID**		
1.	–	0's	→	Network ID
2.	–	1's	→	Direct Broadcast Address (DBA)
3.	1's	1's	→	Limited Broadcast Address (LBA)
4.	0's	–	→	Host with in the Network
5.	1's	0's	→	Network Mask or Subnet Mask

1.7 Multicast communication

Transmitting a packet from one computer to many computers (0 or more) is called Multicast communication.

1.8 CIDR Rules

1. All the IP Address in the Block must be contiguous.
2. Block size must be a power of 2.
3. First IP address of the block must be divisible by size of the block.

1.9 Supernetting

The process of combining two or more network to get a single network is called as supernetting.

1.10 Advantage of Supernetting

1. Super netting Reduce Routing table entry.
2. Router will take less time for processing the packet.
3. It improve flexibility of IP Address Allotment i.e. If someone required 500 Address, then no need to purchase class B network we can combine two class C network.

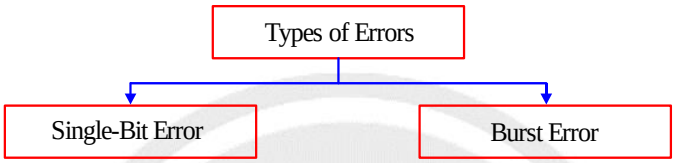
1.11 Rules of Supernetting

1. Network ID must be contiguous.
2. Size of the Network must be same and number of Network must be a power of 2.
3. First Network ID must be divisible by size of the supernet.

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2 ERROR CONTROL

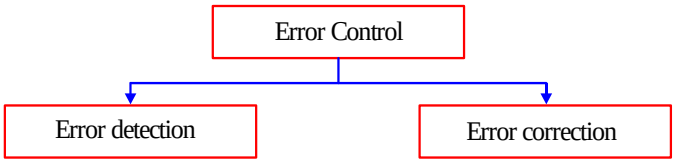
2.1 Error Control



Note:

- The number of corrupted bits or affected bits depends on the data rate and duration of noise.
- The number of corrupted bits or affected bits =Data rate * Noise duration.
- Burst error is more likely to occur than a single bit error.
- Error correction is more difficult than error detection.

	Error Detection	Error Correction
1.	Once noticed error simply discard.	Capability of correcting error.
2.	Ask for retransmission.	Does not required retransmission.



1.	Simple Parity	1.	Hamming code
2.	2D parity		
3.	Check sum		
4.	CRC		

2.1.1 Hamming distance

Hamming distance between two Binary string of same size is the number of differences between corresponding bits.
Hamming distance between two Binary string is denoted by d(x, y)

d(000, 011) = 2

d(100, 011)=3
d(10101, 11110)=3

Hamming distance can easily be found if we apply XOR operation (⊕) on the two words and count the Number of 1's in the result.

2.1.2 Minimum Hamming distance

In a set of codewords, the minimum Hamming distance is the smallest Hamming distance between all possible pairs of code words.

Valid code word		
0	1	0 (a)
1	0	1 (b)
1	1	0 (c)
0	0	1 (d)

d (a, b) = 3
d (a, c) = 1
d (a, d) = 2
d (b, c) = 2
d (b, d) = 1
d (c, d) = 3

Minimum Hamming distance = 1

2.1.3 Minimum Hamming distance for Error detection

To detect 'd' bit error minimum hamming distance required =d+1

2.1.4 Min. Hamming distance For Error Correction

To Correct 'd' bit error minimum hamming distance required =2d+1

2.2 Simple Parity Check Code

Simple parity

In the Simple parity concept one extra bit (parity bit) is added to each dataword.
Simple parity check can detect all single bit error .
Simple parity check can not detect an even number of errors.
Simple parity check can detect an odd number of errors .

2.3 2D Parity Check Code

2D parity

Two dimensional parity check can detect and correct all single bit error and detect two or three bit error that occur any where in the matrix.
However only some pattern with four or more Errors can be detected.
In a 2D-parity check code, the information bits are organized in a matrix consisting of rows and columns.
For each row and each column one parity check bits is calculated.

2.4 CRC

Length of the dataword = n
Length of the divisor = k
Append (k-1) Zero's to the original message
Perform modulo 2 division
Remainder of division = CRC
Codeword = dataword with Appended (k-1) Zero's + CRC

Note:

1. CRC must be (k-1) bits.
2. If the generator has more than one term and coefficient of x^0 is 1, all single bit error can be detected.
3. If a generator can't divide $x^t + 1$ (t between 0 and n – 1) then all isolated Double error can be detected.
4. The generator that contains a Factor of x + 1 can detect all odd numbered errors.

2.4.1 A good polynomial generator needs to have the following characteristics

1. It should have at least two terms.
2. The coefficient of the term x^0 should be 1.
3. It should not divide $x^t + 1$, for t between 2 and n – 1.
4. It should have the factor x + 1.

2.5 Hamming Code

1. Hamming code is used for error correction.
2. Hamming code can correct 1 bit error only.
3. Hamming code can detect upto 2 bit error.

   m = Massege bits

   r = redundant bits or Check bits or parity bits or extra bits

   n = m + r (n = codeword)

According to the hamming code, number of redundant bits

$$\boxed{m + r + 1 \leq 2^r}$$

where r = lower limit

❑❑❑

3 FLOW CONTROL

3.1 Delay in Computer Network

(1) Transmission Delay (T_d)
(2) Propagation Delay (P_d)
(3) Queuing Delay (Q_d)
(4) Processing Delay (P_{rd})

3.2 Transmission Delay

Amount of time taken to transfer a packet on to the outgoing link is called as Transmission delay.

$$\text{Transmission delay}\,(T_d) = \frac{\text{Length of packet}}{\text{Bandwidth}}$$

$$T_d = \frac{L}{B}$$

3.3 Propagation Delay

Amount of time taken to reach a packet from one (sender) point to another (receiver) point is called as propagation delay.

$$\text{Propagation delay}\,(P_d) = \frac{\text{distance}}{\text{velocity}}$$

$$P_d = \frac{d}{v}$$

3.4 Stop wait Protocol

3.4.1 Sender Side Rule

Rule 1 : Sender can send one data packet at a time.
Rule 2 : Sender can send the next data packet only after receiving the ACK of the previous packet.

3.4.2 Receiver Side Rule

Rule 1: Receiver will receive and consume the data packet.
Rule 2 : After consuming the data packet, Ack need to be sent.

3.5 Efficiency OR Line utilization OR Link utilization OR Sender utilization

$$\text{Efficiency} = \frac{\text{Useful time}}{\text{Total time}}$$

$$\text{Efficiency} = \frac{T_d(\text{frame})}{T_d(\text{frame}) + 2*P_d + Q_d + P_{rd} + T_d(\text{ACK})}$$

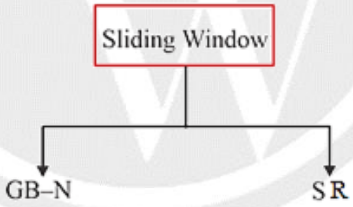
3.6 Throughput Or Effective Bandwidth Or Bandwidth Utilization Or Maximum Data Rate Possible

$$\text{Throughput} = \frac{\text{Length of the Frame}}{\text{Total time}}$$

$$\boxed{\text{Throughput} = \frac{L}{T_d(\text{frame}) + 2*P_d + Q_d + P_{rd} + T_d(\text{ACK})}} \quad \text{OR} \quad \boxed{\text{Throughput} = \eta * B}$$

3.7 Sliding Window

In the sliding window concept instead of sending one packet and wait for the acknowledgement, we send 'w' packet and wait for the Acknowledgement. Where 'w' is the sender window size.



3.7.1 GB–N(N>1)

1. In the GB – N the sender window size is N itself.
2. In the GB-N the receiver window size is equal to one always.

Note:

(1) Out of order packet is not received by Receiver.

(2) Timer is maintained only for the first frame (Rightmost) in window because if its timer expires then sender assume that rest of the frame are also not received by receiver (because out of order delivery is rejected).

Note:

GB–N uses cumulative Acknowledgement and Acknowledgement number defines the number of the next expected frame.

Ack timer < Time out timer

Window Receiver (W_R) size:

In the GB-N the window receiver size is equal to one always irrespective of window sender size (Wr=1).

Window Sender (W_S) Size:

Window sender size is calculated based on the following formula:

$Ws + W_R \leq$ Available Sequence Number

$Ws + 1 \leq$ Available Sequence Number

$Ws \leq$ Available Sequence Number – 1

3.7.2 Efficiency and Throughput in GBN

$$\text{Efficiency} = \frac{\text{Useful time}}{\text{Total time}}$$

$$\boxed{\text{Efficiency} = \frac{N*T_d(frame)}{T_d(frame)+2*P_d+Q_d+P_{rd}+T_d(ACk)}}$$

$$\text{Throughput} = \frac{N*\text{Length of Frame}}{\text{Total time}}$$

$$\boxed{\text{Throughput} = \frac{N*\text{Length of Frame}}{T_d(frame)+2*P_d+Q_d+P_{rd}+T_d(ACk)}} \quad \text{OR} \quad \boxed{\text{Throughput} = \eta * B}$$

3.8 Selective Repeat ARQ

3.8.1 Selective Repeat/Selective Reject ARQ

(1) In SR Protocol window sender size is equal to window receiver size ($W_S = W_R$).

(2) SR Protocol uses independent acknowledgement, and acknowledgement number defines number of error free packet received.

(3) SR receiver can receive out of order packet but packets are delivered to upper layer in sorted order.

(4) In SR protocol searching and sorting logic is required. Searching is done by sender and sorting is done by receiver.

(5) Timer is maintained for each and every frame in the window at sender side.

(6) For 1st out of order delivery or if packet received is corrupted then Negative acknowledgment (NACK) for respective packet is sent by receiver to sender.

(7) When sender receive NACK 3 then it will search in the window for packet 3 & immediately packet 3 is retransmitted even though its timer is not expired.

3.8.2 Efficiency and Throughput of SR

$$\text{Efficiency} = \frac{\text{Useful time}}{\text{Total time}}$$

$$\boxed{\text{Efficiency} = \frac{Ws \times T_d(frame)}{T_d(frame)+2*P_d+Q_d+P_{rd}+T_d(ACK)}}$$

$$\boxed{\text{Throughput} = \eta * B}$$

$$\text{Throughput} = \frac{W_s * \text{Length of the frame}}{\text{Total time}}$$

$$\boxed{\text{Throughput} = \frac{W_s * \text{Length of the frame}}{T_d(frame)+2*P_d+Q_d+P_{rd}+T_d(ACK)}}$$

3.9 Comparison among Stop and Wait, GBN and SR protocols

	Stop & wait	GBN	SR
Efficiency	$\eta = \frac{\text{Useful time}}{\text{Total time}}$ or $\eta = \frac{T_d(frame)}{\text{Total time}}$	$\eta = \frac{\text{Useful time}}{\text{Total time}}$ or $\eta = \frac{N*T_d(frame)}{\text{Total time}}$	$\eta = \frac{\text{Useful time}}{\text{Total time}}$ or $\eta = \frac{W_s*T_d(frame)}{\text{Total time}}$
Throughput	$\frac{\text{Length of frame}}{\text{Total time}}$ or $\eta * B$	$\frac{N*\text{Length of the frame}}{\text{Total time}}$ or $\eta * B$	$\frac{W_s*\text{Length of the frame}}{\text{Total time}}$ or $\eta * B$
Buffer	1 + 1	N + 1	N + N
Seq No.	2	N + 1	2N
Seq. No. = K bit		$W_s = 2^K - 1$, $W_R = 1$	$W_S = 2^{K-1}$, $W_R = 2^{K-1}$

$$\boxed{\text{RTT or Total Time} = T_d(frame) + 2*P_d + T_d(ACK) + P_{rd} + Q_d}$$

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4 IPv4 HEADER

4.1 IPv4 Header

VER (4 bits)	HL (4 bits)	Services (8 bits)	Total Length (16 bits)
Identification number (16 bits)		Flags (3 bits)	Fragment offset (13 bits)
Time to Live (8 bits)		Protocol (8 bits)	Header checksum (16 bits)
Source IP Address (32 bits)			
Destination IP Address (32 bits)			
Option (0-40 bytes)			

4.1.1 Version (4 Bit)

It is used to indicate IPv4 or IPv6.

4.1.2 Header Length(4 Bit)

Header length is a 4 bit field that contains the length of header.

Minimum Header size is 20 byte.

Maximum Header size is 60 byte.

4.2 Services [8 bit]

In this Interpretation the first 3 bit are called precedence bit (Priority bit) and Next 4 bit are called types of services bits and last bit is Not used.

4.2.1 Priority

It is a 3-bit subfield ranging from 0 to 7 (000 to 111 in binary). Priority field is needed if a router is congested and need to discard some datagram, those datagrams which have the lowest priority are discarded first.

4.2.2 Types of Services

It is a 4 bit subfield. Each bit having a special meaning, although a bit can be 0 or 1. One and only one of the bits can have the value 1 in each datagram.

4.3 Total length (16 bits)

Total length = Data + Header

4.4 Identification Number (16 bits)

1. Each datagram is associated with a sequence number is called as datagram number or identification number.
2. It is used to identify all the fragment of same datagram.
3. All the fragment of same datagram will have the same identification number.

4.5 Flags

It is the 3 bit field shown in the figure.

X	D F	M F
Not Used	Don't Fragment	More Fragment

4.6 Fragment offset (13 bits)

Fragment offset indicate no of data byte ahead of this fragment in that particular packet.

4.7 TTL (8 bits)

1. TTL is used to avoid infinite looping.
2. TTL field is used to control the maximum number of hops visited by datagram.
3. When a source host sends a datagram, it stores a number in this field. Each router that process the datagram decrements this number by one. If TTL field reaches zero before the datagram arrives at its destination, then the datagram is discarded and an ICMP message is sent back to sender.

4.8 Protocol (8 bits)

1. This 8 bits field tell us which protocol is encapsulated in the IP packet.
2. At the time of traffic, some packet must be discarded. In this case it will be advantageous to know which protocol data it contains.
3. The order in which router eliminate the datagram from buffer is-
   ICMP>IGMP>UDP>TCP
   (01) (02) (17) (06)

4.9 Header Checksum

1. It is calculated only for header part not the data because rest of the component in packet already covered by TCP checksum.
2. Header checksum is calculated at each and every Router because IP Header might be change when packet is moving from one router to another.
3. Every router makes one modification i.e. TTL so Header checksum is calculated at every Router.
4. Fragment offset, MF, Total length, option all may be changed at a Router.

4.10 Source Address (32 bits)

This 32 bit defines the IPv4 address of source. This field remain unchanged during the time the IPv4 datagram travel from the source Host to destination Host.

4.11 Destination Address (32 bits)

This 32 bits Field defines the IPv4 address of the destination. This field remain unchanged during the time the IPv4 datagram travel from source host to destination host.

4.12 Option

The Header of IPv4 data gram is made of two parts a fixed part and a variable part. The fixed part is 20 bytes long and variable part that can be maximum of 40 bytes.

There are 5 options

1. Strict source Routing
2. Loose source Routing
3. Record Routing
4. Time stamp
5. Padding

4.12.1 Strict Source Routing

A strict source routing is used by the source to predetermine a route for data gram as it travel through the internet.

4.12.2 Loose Source Routing

A loose source route option is similar to strict source route but it is less rigid. Each router in the list must visited, but the data gram can visit other router as well.

4.12.3 Record Routing

A record route option is used to record the internet routers that handle the data gram. It can list up to 9 router Address.
All the Router are supposed to record their IP Address on their IP packets.

Note:

First 16 bits (2 byte) are reserved for option type (8 bit) and length (8 bit). Out of 40 bytes only 38 bytes are remaining for storing IPv4 addresses. In 38 bytes we can store 9 IPv4 addresses as each IPv4 address is of 4 byte

4.12.4 Time Stamp

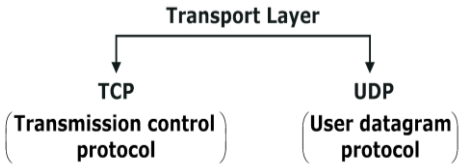
It is used to find out delays at each router. Every router should record incoming time and outgoing time.

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5 TCP & UDP

5.1 Introduction

Transport Layer can be connection oriented or connection less.



1. TCP is reliable process to process delivery of entire message.
2. TCP is a connection oriented.
3. TCP connection are full duplex and point to point.
4. TCP connection has 3 phases
   i. Connection establishment
   ii. Data transfer
   iii. Connection termination
5. Each TCP connection is identified uniquely by Source Port + Destination Port + Source IP + Destination IP.
6. Each TCP connection is associated with Four window.
7. TCP uses three way “Handshake” to establish TCP connection.
8. TCP is not useful for Broadcasting and Multicasting.
9. TCP Header size is 20 bytes but if option (40 bytes) is added it will become 60 bytes.
10. TCP provide end to end error control and flow control.
11. Data will be received at the destination in order.
12. Data may arrive out of order and be temporarily stored by receiving TCP, but TCP guarantee that no out of order data delivered to the process.

5.2 TCP header

<table>
<tr><td colspan="8">Source Port (16 bits)</td><td>Destination Port (16 bits)</td></tr>
<tr><td colspan="9">Sequence number (32 bits)</td></tr>
<tr><td colspan="9">Acknowledgement number (32 bits)</td></tr>
<tr><td>Header Length (4 bits)</td><td>Reserved bits (6 bits)</td><td>URB</td><td>ACK</td><td>PSH</td><td>RST</td><td>SYN</td><td>FIN</td><td>Window Size (Advertisement Window) (16 bits)</td></tr>
<tr><td colspan="8">Check sum (16 bits)</td><td>Urgent Pointer (16 bits)</td></tr>
<tr><td colspan="9">Options
(0 - 40 bytes)</td></tr>
</table>

Port Number	Name
0 – 1023	Well known port No.
1024 – 49151	Registered Port No.
49152 – 65535	Dynamic Port No.

$$\text{Wrap Around time (WAT)} = \frac{\text{Total sequence No.}}{\text{Bandwidth}\left[\text{Bytes / Sec}\right]}$$

- Minimum sequence number required to Avoid wrap Around time with in Life time = B × LT
- Minimum number of bits required to Avoid wrap Around time with in LT = $\lceil \log_2 B * LT \rceil$

SYN = 1 → Consume one sequence number.

Ack = 1 → Consume zero sequence number.

FIN = 1 → Consume one sequence number.

1 Data byte → Consume one sequence number.

SYN	Ack	Meaning
1	0	request
1	1	reply
0	1	Ack
0	0	Data

Time out timer in TCP

Basic Algorithm	Jacobson's Algorithm
Time Out Timer = 2 * RTT Next Round Trip Time (NRTT) = α(IRTT) + (1 – α)ARTT $0 \le \alpha \le 1$	Time Out Timer = 4 * ID + RTT Next Round Trip Time (NRTT) = α (IRTT) + (1 – α) ARTT $0 \le \alpha \le 1$ Actual Deviation (AD) = |IRTT – ARTT| Next Deviation (ND) =α (ID) + (1 – α) AD

5.3 Congestion Control

1. $W_s = \min (W_C, W_R)$

Slow start	Congestion Avoidance	Congestion Detection
1. If ACK Arrives $W_c = W_c + 1$	1. IF ACK Arrives $W_c = W_c + \frac{1}{W_c}$	1. Time out
2. After one RTT $W_c = 2 * W_C$	2. After one RTT $W_c = W_c + 1$	2. 3 duplicate ACK

5.4 Token Bucket

- Token bucket algorithm allows ideal hosts to accumulate credit for the future in the form of tokens.
- In regular interval, tokens are thrown into the bucket.
- Bucket has a maximum capacity.
- If there is a ready packet a token is removed from bucket and packet is sent.
- If there is no token in the bucket, the packet cannot be sent.

Let the capacity of token bucket is 'C' token and token enter into the bucket at rate of 'r' tokens per second.
The maximum number of packet that can be enter into the network during the time interval 't' is

Maximum number of packet = $C + rt$

Maximum average rate for token bucket $M = \frac{C + rt}{t}$

$$Mt = C + rt$$
$$Mt - rt = C$$
$$(M - r)t = C$$
$$t = \frac{C}{M - r}$$

C → token Bucket capacity
r → Token Arrival rate

5.5 UDP

1. UDP is message oriented connection less Datagram protocol.
2. It is unreliable Transport protocol.
3. It does not provide Flow control and Error control & congestion control.
4. It does not add anything to the services except process to process delivery of data.
5. Header is simple and fixed in size i.e. 8 byte.

UDP Header

Source port (16 bit)	Destination port (16 bit)
Length (16 bit)	Checksum (16 bit)

Note:
Unlike TCP, the checksum calculation is not mandatory in UDP. No error control or flow control is provided by UDP. Hence UDP depends on IP and ICMP for error reporting.

5.5.1 Optional inclusion of checksum

The sender of UDP packet can choose not to calculate the checksum. In this case the checksum field is filled with all 0's before being sent.

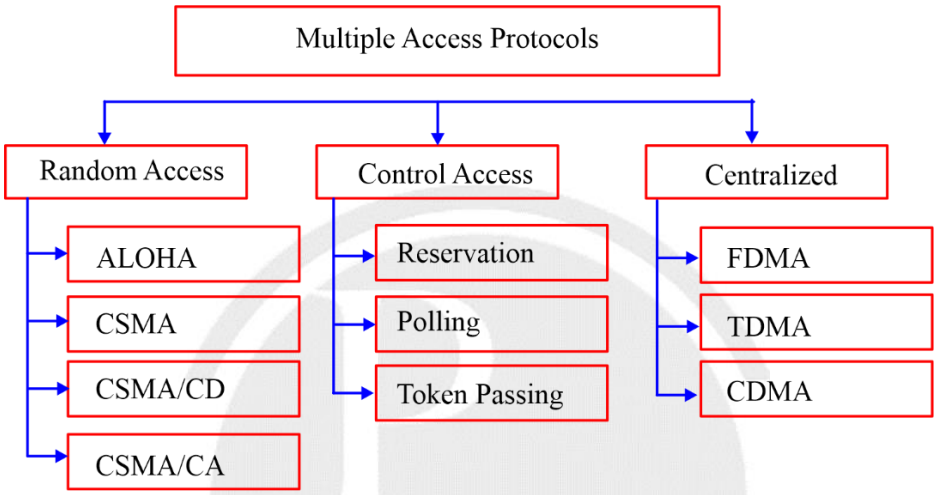
5.5.2 Need of UDP

[1] The application that required one request one reply. TCP is not suitable hence we use UDP.
[2] Application that required constant dataflow TCP is not suitable hence we use UDP.
[3] Application that required multimedia data transfer we cannot use TCP hence we use UDP.
[4] Application that required fastness and then reliability TCP is not suitable hence we use UDP.
[5] UDP used for management process such as SNMP (simple network management protocol).
[6] UDP is used for some route updating protocol such as RIP.
[7] For broadcasting & multicasting application TCP is no suitable hence we use UDP.
[8] UDP is normally used for interactive real time applications.
[9] UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer protocol(TFTP) process include flow and error control. It can easily use UDP

TCP	**UDP**
Connection-oriented	Connectionless
Reliability in delivery of message	Not reliable
Sequence Number.	No sequence number.
ACK number.	No ACK number.
Overhead is high(Header size 20-60 Bytes)	overhead is less(Header size = 8 Bytes)
Keep track of order (sequence)	No order
Protocols: HTTP, FTP, SMTP, POP	Protocol: DNS, SNMP, TFTP, DHCP, All real time protocol

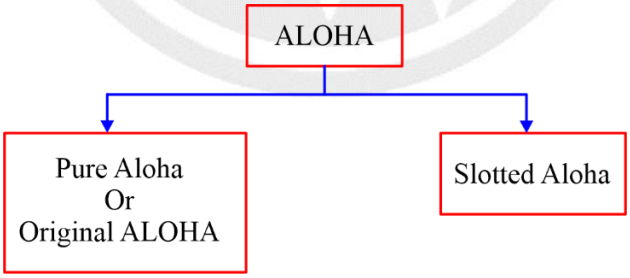
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6 MEDIUM ACESS CONTROL[MAC]



6.1 Aloha

Aloha was developed at university of Hawaii in 1970's.
It was designed for wireless LAN but it can be used in any shared medium.
Each station sends equal size frame.



Pure Aloha	Slotted Aloha
Any station transmits the data at any time.	Any station can transmit the data at the beginning of any time slot.
Vulnerable time in which collision may occur = 2 * T_f (T_f - Transmission time for single frame)	Vulnerable time in which collision may occur = T_f
Throughput of pure aloha = $G * e^{-2G}$	Throughput of slotted Aloha = $G * e^{-G}$
Maximum throughput s_{max} = 18.4 % (When G = 1/2)	Maximum throughput s_{max} = 36.8 % (When G = 1)
The main advantage of pure aloha is it simplicity in implementation	The main advantage of slotted aloha is that it reduces the number of collisions to Half and double the throughput of pure aloha

6.2 CSMA (Carrier Sense multiple access)

CSMA requires that each station, first sense the carrier before transmitting the data.
Vulnerable time for CSMA = Propagation time.

6.3 Persistence methods in CSSMA

Persistent
Non-persistent
P-persistent

6.4 CSMA/CD (Carrier Sense multiple access/Collision Detection)

1. Minimum size of frame to detect the collision in Ethernet (CSMA/CD)
   $T_d \geq 2 * P_d + T_d$(Jam signal)
2. Backoff Algorithm
   Waiting time = K * Slot duration
   = K * RTT
   = K * 2 * P_d

   K is any random number in between 0 to 2^n -1.
   n is collision number (Collision number is respect to data packet).

3. Efficiency in Ethernet (CSMA/CD)

$$\eta = \frac{1}{1+6.44a} \quad \text{or} \quad \eta = \frac{\text{useful time}}{\text{Total time}} = \frac{T_d}{\text{Collision time}+T_d+P_d}$$

4. $P(1-P)^{N-1}$ → Probability of success for single station(Throughput of Host)

$NP(1 - P)^{N-1}$ → Probability of success for any station among all stations [Throughput of channel]

5. Ethernet Frame Structure:

Preamble	SFD	DA	SA	Length Of Data	Data	CRC
7B	1B	6B	6B	2B	(46B – 1500B)	4B

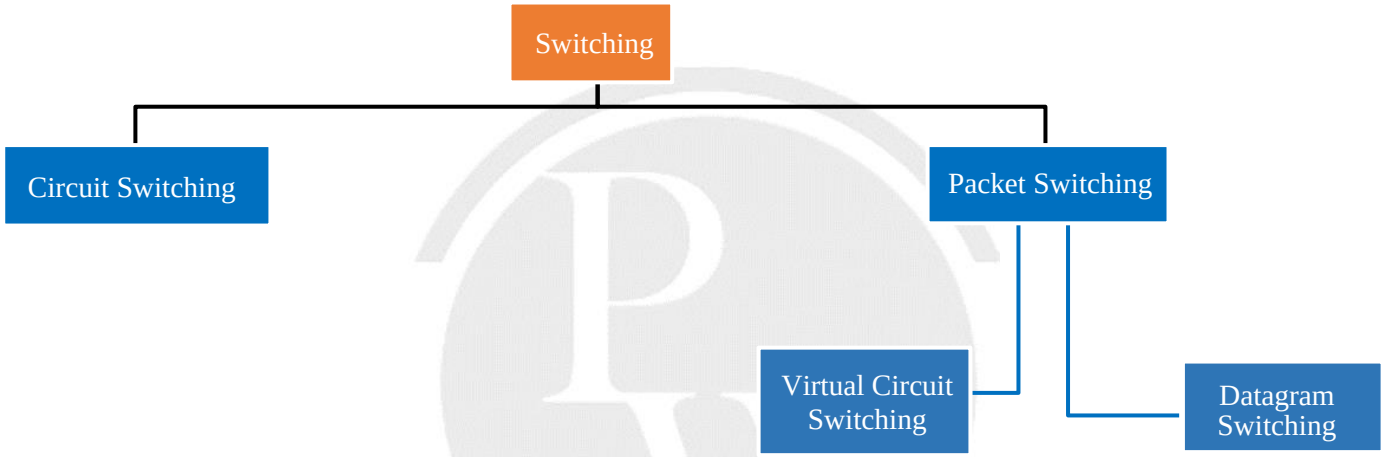
PL DLL DLL

Ethernet uses Manchester encoding technique for converting data bits into signal.
(Baud rate = 2 ∗ bit rate)
Bit rate = 1/2 baud rate

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7 ROUTING ALGORITHMS, SWITCHING & IP SUPPORT PROTOCOL

7.1 Switching



Circuit Switching	Packet switching
(I) It has three phases-connection establishment, Data transfer and Connection termination.	It has only one phase-Data transfer
(2) Physical path between source and destination	No physical path
(3) All packets use same path	Packet may follow different path (travel independently)
(4) Reserves the entire bandwidth in advance	Does not reserve
(5) Bandwidth wastage	No Bandwidth wastage
(6) No store and forward transmission	Supports store and forward transmission
(7) Congestion can happen during connection establishment phase	Congestion can happen during data transfer phase

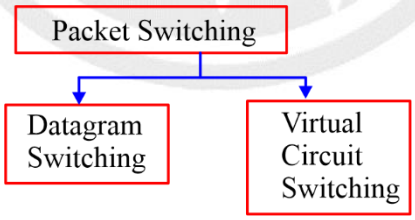
(8) It is reliable	Not reliable
(9) Better for sending large messages	Better for sending small messages
(10) Not fault tolerant technique	Fault tolerant technique
(11) Circuit switching is implemented at physical layer.	Packet switching is implemented at network layer
Total time= Setup time + T_d + P_d + Tear down time $TT=S+\frac{L}{B}+X.\frac{d}{V}+T$	For X Hop and N packet Total time= $X[T_d + P_d] + X-1[P_{rd} + Q_d] + N-1(T_d)$

7.2 Generalized Formula for optimal packet size(P)

M = Message size
h = Header size
p = Payload/Packet data size
No. of Hops = X

$p = \sqrt{\frac{Mh}{X-1}}$

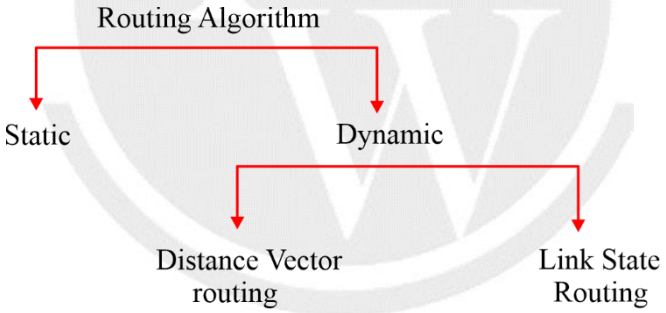
So optimum packet size P = p + h



Datagram Switching	Virtual circuit switching
(1) It is a connection less service.	It is connection-oriented service.
(2) All the packets may follow different path.	All the packet follows the same dedicated path.
(3) Data may appear out of order at the destination since the packets may follow different paths.	Data appears in order at the destination since all the packets take the same dedicated path.
(4) It is not reliable since packets may be discarded.	It is highly reliable.

(5) Cheap	Costly
(6) No resource reservation is done.	First packet reserves the resources (CPU, bandwidth and buffer) for the subsequent packets.
(7) All the packets require a global header which contains full information about the destination.	Only first packet requires a global header which identifies the path from one end to another end. All the following packets require a local header which identifies the path from hop to hop.
(8) The packets may be discarded at intermediate switches if sufficient resources are not available to process the packets.	The packets are never discarded at intermediate switches and immediately forwarded since resources are reserved for them.
(9) IP Networks uses datagram switching.	ATM (Asynchronous Transfer Mode) uses virtual circuit switching.
(10) Datagram switching is normally implemented at network layer.	Virtual circuit switching is normally implemented at data link layer.

7.3 Routing Algorithm



Distance vector Routing	Link state Routing
1. 1980's	1. 1990's
2. Bandwidth required is very less because we sent only distance vector packet.	2. Band width required is high because we sent entire link state packet
3. Local knowledge	3. Global knowledge
4. Bellman Ford algorithm	4. Dijkstra algorithm
5. Traffic is very less	5. Traffic is very high

6. Convergence is very low	6. Convergence is faster
7. Count to infinity Problem	7. No problem of count to infinity
8. Persistent Loops	8. Transient Loops
9. RIP	9. OSPF

The maximum Hop count allowed For RIP is 15 and Hop count of 16 is considered as Destination unreachable.

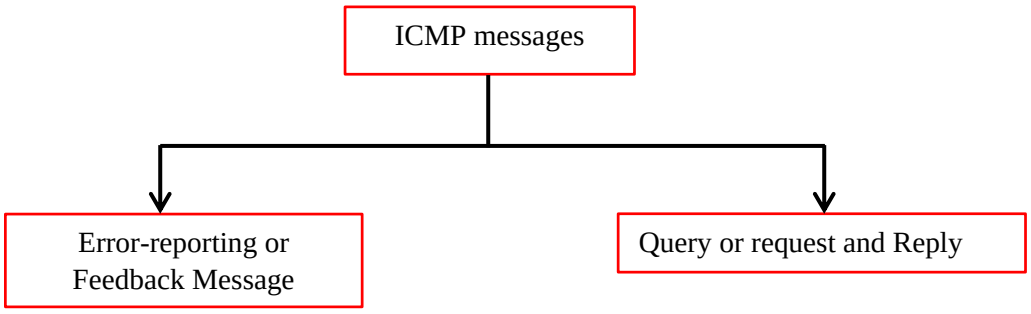
Note:
RIP uses UDP as its transport protocol with the port number – 530

7.4 IP Support Protocol

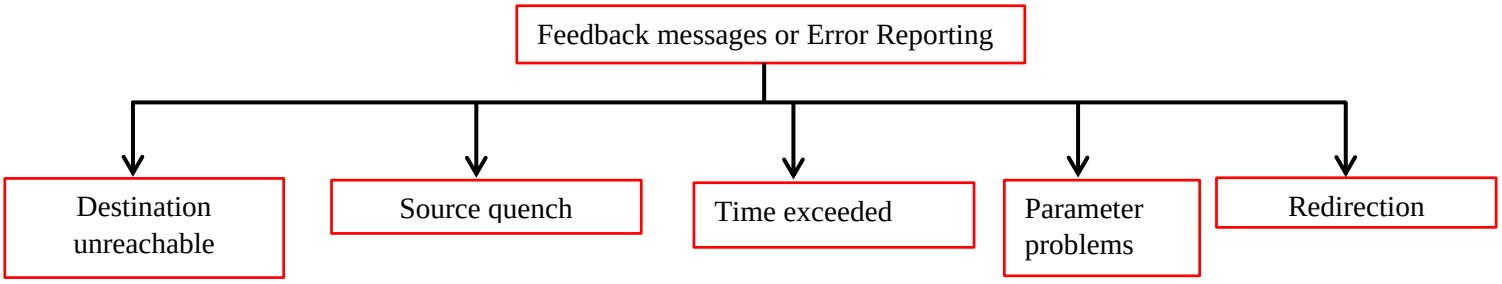
7.4.1 ARP

1. Address Resolution Protocol(ARP) is used to find the MAC(Media Access Control) address of a device from its IP address.
2. **ARP request:** ARP request is broadcasting
3. **ARP response/reply:** ARP reply is unicasting.

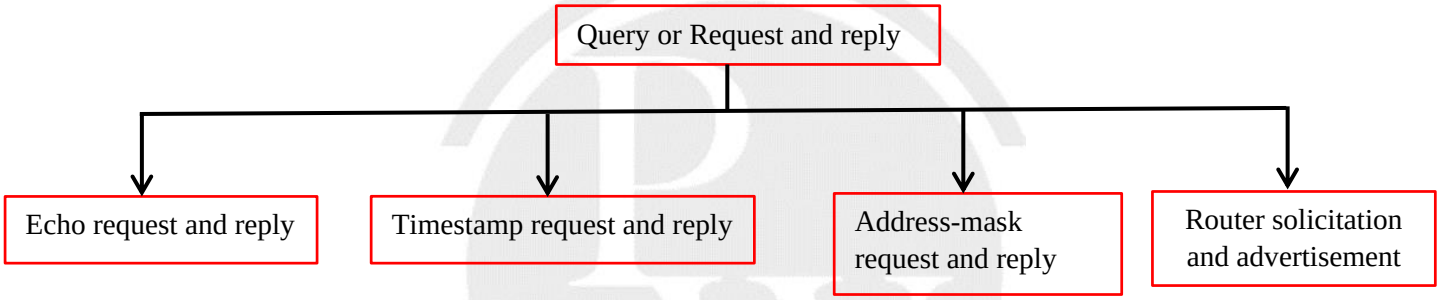
7.4.2 Internet Control Message Protocol (ICMP)



7.4.3 Internet Control Message Protocol (ICMP)



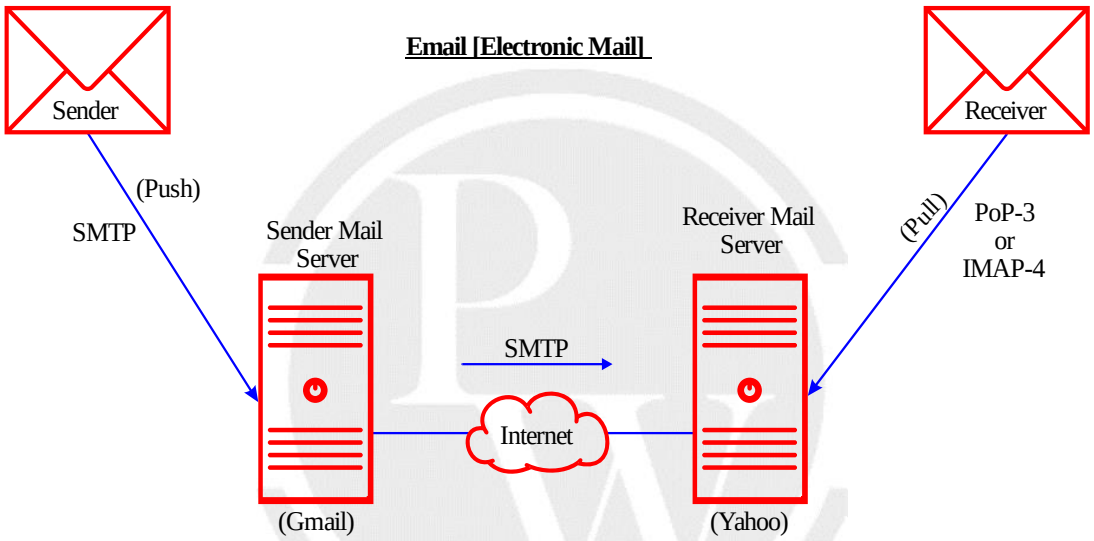
7.4.4 Internet Control Message Protocol (ICMP)



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8 APPLICATION LAYER PROTOCOL

8.1 E-MAIL



SMTP transfer the mail from sender's mail server to receiver's mail server.
While sending the mail ,SMTP is used two times-.

(i) Between the sender and sender's mail server
(ii) Between the sender's mail server and receiver's mail server

8.1.1 To receive or download the email,

Another protocol is needed between the receiver mail server and receiver.
The most commonly used protocols are POP3 and IMAP4.

8.1.2 SMTP(Simple mail transfer protocol)

1. The objective of SMTP is to transfer the email reliably and efficiently.
2. It uses port number-25 at TCP.
3. In SMTP there are two components:
   (i) User Agent (UA)
   (ii) Mail transfer Agent (MTA)
4. User Agent prepares the message, creates the envelope and put the message in the envelope.
5. Mail transfer Agent transfer the mail across the internet i.e. Actual mail transfer is done through MTA.
6. To send mail, system must have a client MTA and to receive the mail. it must have a server MTA.

7. SMTP is text based protocol.
8. With the help of SMTP & POP we can send only text messages.
9. SMTP can only handle the message containing 7 bit ASCII text.
10. SMTP cannot transfer other types of data like images, video, audio, etc.
11. SMTP cannot transfer binary files or executable files.
12. SMTP cannot transfer the text data for the language other than English (such as French, Japanese, and Chinese etc.).
13. Only SMTP is not sufficient to send binary files or to send videos or audio so we require MIME (Multipurpose Internet mail extension).
14. MIME is a supplementary protocol that allows non-ASCII data to send through SMTP.
15. MIME is a set of software function that transforms non-ASCII data to ASCII data or viceversa.
16. MIME is used to convert non text data to text data and text data to non text data.
17. SMTP is stateless protocol. It does not maintain any information of user. If an e-mail is asked to be sent twice, then server resends it without saying that e-mail has already been sent.
18. SMTP is a connection-oriented protocol.
19. SMTP uses persistent TCP connections, so it can send multiple e-mail at once.
20. SMTP is an “In-Band” protocol.
21. SMTP is used for Push the e-mail.
22. SMTP Pushes the mail from client to server on other hand, It needs a pull protocol(Download).
23. POP3 and IMAP4 are used for Pulling the e-mail.

8.2 POP3(Post office Protocol version-3)

1. It is a message access protocol.
2. It is a pull protocol.
3. POP3 uses port number-110 at TCP.
4. POP3 is a connection-oriented protocol.
5. POP3 uses persistent TCP connection.
6. POP3 is a stateful protocol.
7. POP3 is an “In-Band” protocol.
8. POP3 does not allow users to partially check the content of the mail before downloading.
9. POP3 does not allow user to organize the mail on the mail server.

8.2.1 IMAP-4(Internet Mail Access Protocol version-4)

1. IMAP-4 is similar to POP3 but it has more features. IMAP-4 is more powerful and more complex.
2. IMAP-4 provides the following extra functions.
3. A user checks the email header prior to downloading.
4. A user can search the content of the email for a specific string of characters prior to downloading.
5. A user can partially download the email.

6. A user can create, delete, or rename the mail box on the mail server.
7. A user can create a hierarchy of mailbox in a folder for email storage.

8.2.2 Characteristics of IMAP

1. IMAP is a pull protocol.
2. IMAP uses port number-143 at TCP.
3. IMAP is a connection-oriented protocol.
4. IMAP uses persistent TCP connection.
5. IMAP is a tasteful protocol.
6. IMAP is an "In-Band" Protocol.

POP3	IMAP
(1) Mails can only be accessed from a single device.	Mails can be accessed from multiple device.
(2) Download the email from server to a single computer and the copy at the server is deleted.	The email message is stored on the mail server itself.
(3) User cannot organize the mails in the mail box of the mail server.	User can organize mails on the mail server.
(4) It does not allow user to sync emails.	It allows user to sync their emails.
(5) It is unidirectional i.e all the changes made on a device does not effect the content present on the server.	It is bidirectional i.e all the chances made on server or device are made on the other side too.

8.3 File Transfer protocol (FTP)

8.3.1 FTP (File Transfer Protocol)

1. File transfer protocol is a standard internet protocol for transferring files b/w computers over TCP/IP connection.
2. It uses port number - 20 & 21 on TCP.
3. It has two types of connection
   (i) Control connection (port number. - 21)
   (ii) Data connection (port number - 20)
4. Control connection remains connected during the entire interactive FTP session.
5. The data connection is opened and closed for each file transfer activity.
6. When user starts an FTP session, the control connection opens. While the control connection is open, the data connection can be opened and closed multiple times if several files are transferred.
7. FTP uses persistent TCP connections for control information.
8. FTP uses Non-persistent TCP connections for data information.
9. FTP is a connection-oriented protocol.
10. FTP is an "out of band" protocol as data and control information flow over different connection.
11. Some protocols send their request and data in the same TCP connection for this reason they are called as In-bound protocol.
12. HTTP & SMTP are In-Band protocol.
13. FTP is state full protocol.

8.3.2 Transmission mode

FTP can transfer a file across the data connection using one of the following three transmission modes.

(i) Stream mode
(ii) Block mode
(ii) Compressed mode

8.3.3 File Type

FTP can transfer one of the following file types across the data connection:

(i) ASCII file
(ii) EBCDIC file (File format used by IBM)
(ii) Image file

8.3.4 Data structure

FTP can transfer a file across the data connection using one of the following interpretation of the structure of the data:

(i) File structure
(ii) Record structure
(iii) Page structure

8.4 HTTP Protocol

1. HTTP protocol is used mainly to access data on world wide web (www).
2. It is client server protocol using port number - 80 on TCP.
3. HTTP is “In-Band” protocol i.e. both request and data we will send only in one connection.
4. HTTP is a stateless protocol i.e. It does not maintain any information of user.
5. There are two types of HTTP protocol
   (i) Non persistent (1.0)
   (ii) Persistent (1.1)

8.5 Non Persistent (1.0)

In a Non persistent connection one TCP connection is ade for each request/response. This strategy follow the following steps :-

(i) The client opens a TCP connection and sends a request.
(ii) Server sends the response and closes the connection.
(iii) In this strategy , If a file contains link to N-different pictures in different files(all located on same server) the connection must be opened and closed N+1 times.

8.6 Persistent (1.1)

1. In a persistent connection the server leaves the connection open for more request after sending a response.
2. The server closes the connection at the request of client or time out has been reached.

8.6.1 Important Table

SHORT TRICK	DNS	HTTP	SMTP	POP	IMAP	FTP
Stateful/ Stateless	Stateless	Stateless	Stateless	Stateful	Stateful	Stateful
Transport Protocol Used	UDP	TCP	TCP	TCP	TCP	TCP
Connectionless/ Connection oriented	Connection less	Connection less	Connection oriented	Connection oriented	Connection oriented	Connection oriented
Persistent/Non-persistent	Non-persistent	HTTP 1.0 is non persistent HTTP 1.1 is persistent.	Persistent	Persistent	Persistent	Control connection is persistent. Data connection is non-persistent.
Push/Pull	-	-	Push	Pull	Pull	Can't
Port Number Used	53	80	25	110	143	20 for data connection. 21 for control connection.
In band/ Out-of-band	In band	In band	In band	In band	In band	Out-of- band

Application	Port Number.	Transport Protocol
DNS	53	UDP
HTTP	80	TCP
FTP	20 (Data connection) 21(Control connection)	TCP
SMTP	25	TCP
POP	110	TCP
SNMP	161, 162	UDP
TFTP	69	UDP
IMAP	143	TCP
Telnet	23	TCP
DHCP	67 (DHCP Server) 68 (DHCP Client)	UDP

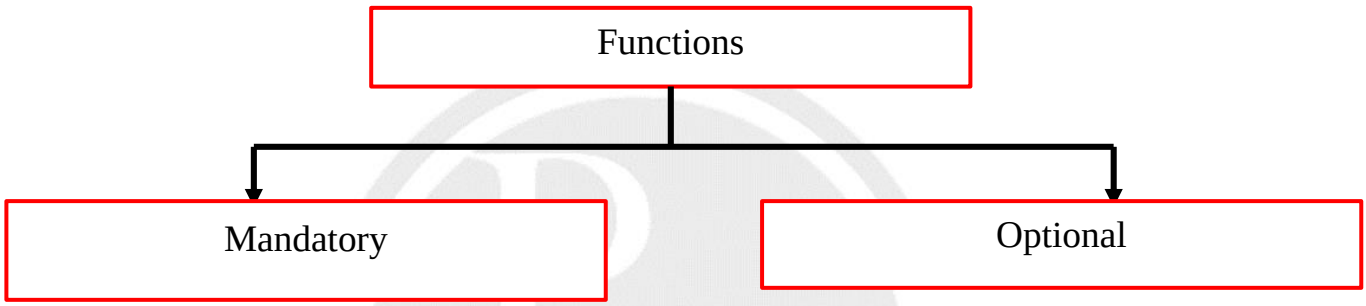
8.6.2 Commands

HTTP	FTP	SMTP
GET	USER	HELO
HEAD	PASS	MAIL FROM
PUT	ACCT	RCPT TO
POST	CWD	DATA
TRACE	REIN	QUIT
DELETE	QUIT	RSET
CONNECT	PORT*	VRFY
OPTIONS	PASV	NOOP
	TYPE	TURN
	MODE	EXPN
	PROMPT	HELP
	STRU	SEND FROM
		SMOL FROM
		SMAL FROM

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9 OSI AND TCP/IP PROTOCOL STACK

9.1 Functions of Computer Network



- Error control
- Flow control
- Access control
- Multiplexing and Demultiplexing
  :
  :

- Encryption & Decryption
- Check pointing
- Routing
  :
  :

9.2 OSI/ISO

OSI : Open systems Interconnection model.
ISO : International Standards Organization. It is a multinational body dedicated to worldwide agreement on international standard.

9.2.1 OSI Mode

- This model has been proposed by ISO.
- An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture (Hardware/Software).
- The purpose is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware & software.
- This model has got 7 separate but related layers.

9.3 Functions of Physical layer

9.3.1 Physical Layer

Physical Layer is responsible for movement of individuals bits from one Hop to next Hop.

9.3.2 Functions of physical Layer

1. It is used to define electrical, mechanical, functional and procedural characteristic of physical link. (physical Link)
   Copper → Electrical signal
   Fiber → Light signal
   Wireless comm. → Electromagnetic signal.

2. It defines transmission mode:
   a. Simplex
   b. Half duplex
   c. Full duplex

3. It defines topology configuration:
   - Bus topology
   - Star topology
   - Mesh Topology
   - Tree Topology

4. It is totally Hardware layer.

5. It defines link configuration:
   i Point to Point Link
   ii Broadcast Link
6. It defines Encoding.
7. Bits Synchronization.
8. Bit rate control.

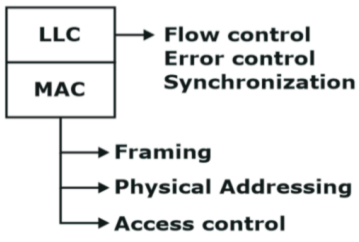
9.4 Data Link Layer

Data link layer is responsible for moving frames From One Hop (Node) to Next Hop (Node).

9.4.1 Function of data link layer

1. Flow control
2. Error control
3. Access control
4. Framing
5. Physical Addressing

Data Link Layer is divided into two parts



9.5 Network Layer

The network layer is responsible for the delivery of individual packet from source to destination.

9.5.1 Function of Network Layer

1. Host to Host connectively
2. Logical Addressing
3. Switching
4. Routing
5. Fragmentation
6. Congestion control:

9.6 Transport Layer

Transport Layer is responsible for process to process delivery. A process is an application program running on a host.

9.6.1 Function of Transport Layer

1. End to end connectively
2. Service point Addressing
3. Flow control
4. Error control
5. Segmentation and Reassembly
6. Congestion control
7. Connection Control
8. Multiplexing and Demultiplexing.

9.7 Session Layer

Session layer also known as **network dialog controller.** It establishes, maintains, synchronizes and terminates the interaction b/w sender and receiver.

9.7.1 Function of Session Layer

1. Authentication & Authorization
2. Check point or synchronization
3. Dialog control

9.8 Presentation Layer

This layer take care of syntax and semantics of the information exchange in between two communicating systems.

9.8.1 Function of Presentation Layer

1. Character translation
2. Encryption/Decryption
3. Compression

9.9 Application Layer

Application Layer is responsible for providing services to users. Users such as:

1. Mail services
2. File sharing
3. File transfer and many more

9.10 TCP/IP Model

This mode has got 5 different layer

Physical : No specific protocol

Data Link : No specific protocol

Network : ARP, RARP, ICMP, IGMP

Transport : TCP, UDP,SCTP

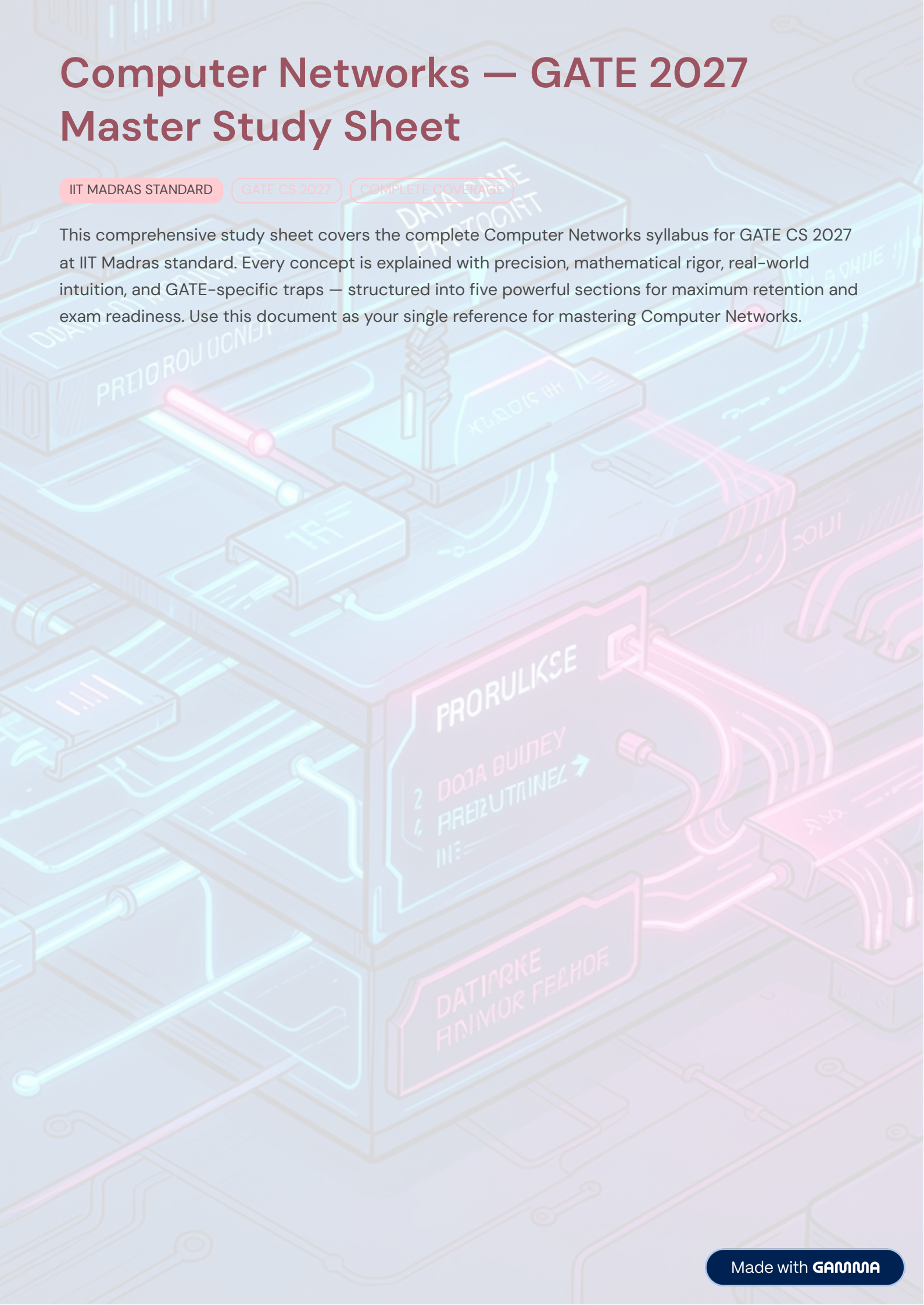
Application : DNS, SMTP, HTTP, FTP, POP,IMAP, SNMP , Telnet etc.

❑❑❑

Computer Networks — GATE 2027 Master Study Sheet

IIT MADRAS STANDARD GATE CS 2027 COMPLETE COVERAGE

This comprehensive study sheet covers the complete Computer Networks syllabus for GATE CS 2027 at IIT Madras standard. Every concept is explained with precision, mathematical rigor, real-world intuition, and GATE-specific traps — structured into five powerful sections for maximum retention and exam readiness. Use this document as your single reference for mastering Computer Networks.



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SECTION 1 — HIGH-YIELD CONCEPT & INTUITION

The Big Picture: Layering, Switching, and Network Architecture

Think of Computer Networks like a **multi-level corporate office building**. Each floor handles a specific job and passes information to the floor above or below. The CEO (Application Layer) never worries about how the mailroom (Physical Layer) delivers packages — each layer trusts the one below it. This is the essence of **layering**: separation of concerns, modularity, and abstraction.

The **OSI Model** has 7 layers (All People Seem To Need Data Processing: Application, Presentation, Session, Transport, Network, Data Link, Physical). The **TCP/IP Model** collapses these into 4 practical layers (Application, Transport, Internet, Network Access). For GATE, remember: OSI is the *theoretical reference*; TCP/IP is what the *Internet actually uses*.

Now consider **switching**. Imagine three ways to travel from Chennai to Delhi:

🚂 Circuit Switching

Like booking an entire train coach. A **dedicated path** is reserved before data flows. No sharing, no congestion — but terribly inefficient if you send nothing. Used in traditional telephony. **Setup delay + Transmission delay**.

📦 Packet Switching

Like sending postcards independently. Each packet finds its own route. **Store-and-forward** at each hop. Efficient but variable delay (jitter). Used in the Internet. Two types: **Datagram** (connectionless) and **Virtual Circuit** (connection-oriented).

🔗 Virtual Circuit

A hybrid — like a pre-booked courier route. Path is **decided at setup**, but bandwidth is shared. Packets follow the same route, arrive in order. Used in ATM, Frame Relay. Combines reliability of circuit with efficiency of packet switching.

Performance metrics you must master: **Bandwidth** (bits/sec — the pipe width), **Latency** (time for first bit to arrive), **Throughput** (actual data rate achieved), **Propagation Delay** = d/v (distance ÷ signal speed), **Transmission Delay** = L/R (packet size ÷ bandwidth), and **RTT** (Round Trip Time = 2 × propagation delay). The **Bandwidth-Delay Product** = R × d/v tells you how many bits can be "in flight" — critical for TCP window sizing.

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SECTION 1 — CONTINUED

Data Link Layer: Framing, Error Detection, and MAC

The **Data Link Layer** (Layer 2) is the neighborhood postman — it ensures frames reach the next hop reliably. Key responsibilities: **Framing** (packaging bits into frames), **Error Detection** (CRC, Parity, Checksum), and **Medium Access Control** (who talks when on a shared medium).

Framing methods: Character Count (simple but error-prone), **Flag Bytes with Byte Stuffing** (HDLC uses 01111110 as flag; escape 0x7D), **Bit Stuffing** (insert 0 after five consecutive 1s in data), and **Physical Layer Coding Violations** (used in Ethernet with Manchester encoding).

Error Detection — Three techniques for GATE: (1) **Parity Check** — detects odd number of bit errors, (2) **Checksum** — used in IP/TCP/UDP, 1's complement addition, (3) **CRC (Cyclic Redundancy Check)** — most powerful, uses polynomial division. For CRC: append r zeros (r = degree of generator polynomial G(x)), divide using XOR, replace zeros with remainder. Receiver divides received frame by G(x); zero remainder = no error. Common GATE generator: **CRC-32** polynomial.

MAC Protocols: In shared media (like old Ethernet hubs), nodes must coordinate. **ALOHA** (pure: transmit anytime, efficiency 18.4%; slotted: transmit at slot boundaries, efficiency 36.8%). **CSMA** (Carrier Sense — listen before talk). **CSMA/CD** (Collision Detection — used in Ethernet; abort on collision, binary exponential backoff). **CSMA/CA** (Collision Avoidance — used in WiFi; RTS/CTS handshake). **Token Passing** (deterministic, no collisions — Token Ring, FDDI).

Ethernet (IEEE 802.3) — Key Facts

Uses **CSMA/CD**. Minimum frame size = 64 bytes (ensures collision detected before transmission ends). Maximum = 1518 bytes. **Manchester Encoding** (self-clocking, 50% efficiency). MAC address = 48 bits (6 hex bytes). **Binary Exponential Backoff:** after k collisions, wait random time in $[0, 2^k - 1]$ slot times. After 16 collisions → abort.

Ethernet Bridging (Transparent Bridge)

A bridge learns MAC addresses by **backward learning** — records source MAC + port on every frame. Uses **Spanning Tree Protocol (STP)** to prevent loops. Forwarding: if destination MAC known → forward to specific port; if unknown → flood to all ports except incoming. Self-learning, plug-and-play.

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SECTION 2 — DEEP-DIVE CORE MECHANICS

Routing Protocols: Distance Vector, Link State, and Shortest Path

Routing is the GPS of networks — finding the best path from source to destination. GATE tests three major paradigms: **Flooding**, **Distance Vector (DV)**, and **Link State (LS)**.

Distance Vector Routing (Bellman-Ford)

Each node maintains a **distance vector** — cost to every other node. Periodically shares entire vector with *only direct neighbors*. Uses **Bellman-Ford equation:**

$$D_x(y) = \min_v \{c(x,v) + D_v(y)\}$$

Where D_x(y) = cost from x to y, c(x,v) = direct link cost. Problem: **Count-to-Infinity** — bad news travels slow. Solutions: **Split Horizon**, **Poison Reverse**, **Triggered Updates**. Protocol: **RIP** (max hop count = 15; ∞ = 16).

Link State Routing (Dijkstra's Algorithm)

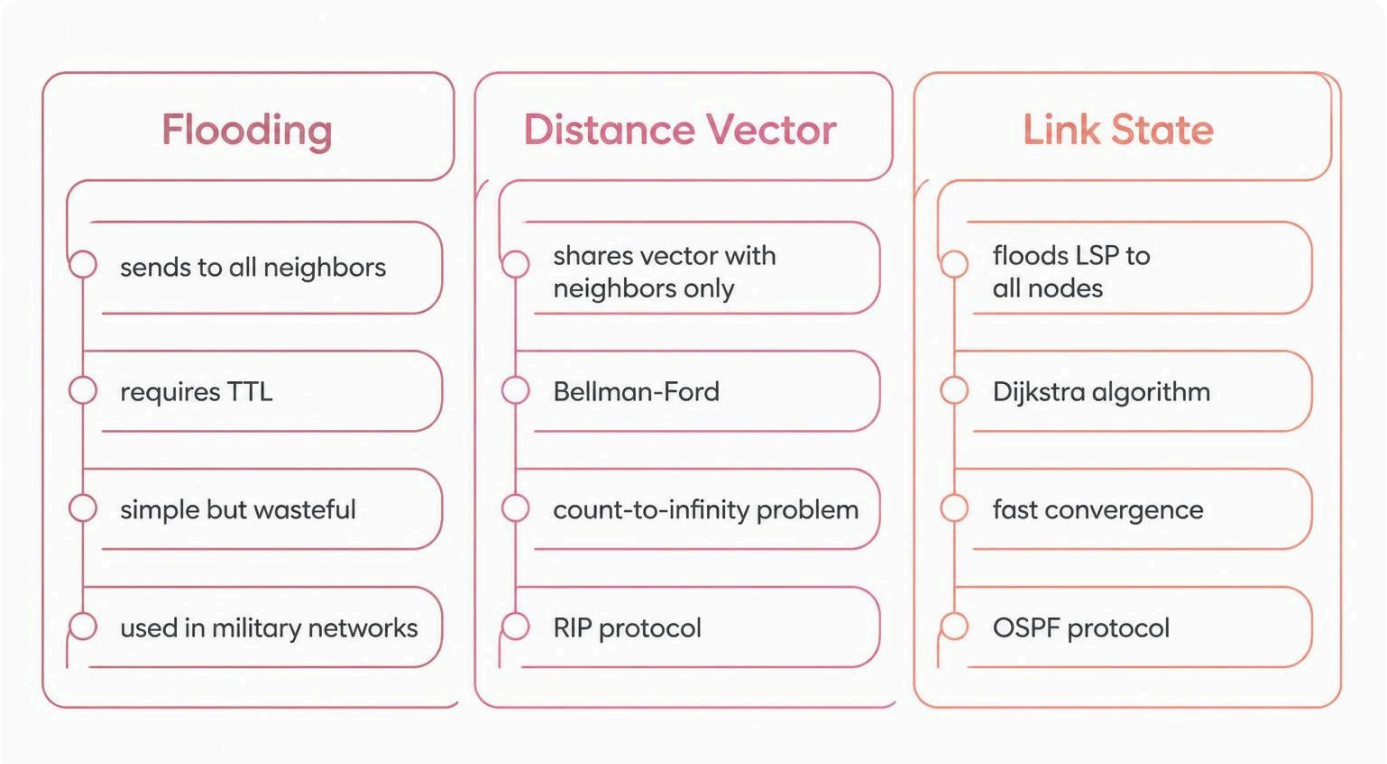
Each node floods **Link State Packets (LSPs)** to *all nodes* in network. Every node builds complete topology map, then runs **Dijkstra's algorithm** locally:

1. Initialize: N' = {source}, D(v) = c(source, v) for direct neighbors, ∞ otherwise
2. Find w ∉ N' with minimum D(w)
3. Add w to N'; update D(v) = min(D(v), D(w) + c(w,v))
4. Repeat until all nodes in N'

Protocol: **OSPF**. Faster convergence, no count-to-infinity, but higher memory/CPU. Uses **Dijkstra complexity O(n²)** or O((n+m)log n) with heap.

Flooding is the simplest: every incoming packet is sent out on all links except the incoming one. Requires **TTL (Time-To-Live)** or sequence numbers to prevent infinite loops. Used in military networks and as a subroutine in link-state routing for LSP distribution.

Comparison for GATE: DV sends less data per update but converges slowly; LS sends more initially but converges fast. DV is vulnerable to routing loops; LS is not. RIP = DV (UDP port 520); OSPF = LS (runs directly over IP, protocol 89).



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SECTION 2 — CONTINUED

IPv4: Fragmentation, CIDR, ARP, DHCP, ICMP, and NAT

IPv4 Fragmentation: When a packet exceeds the **MTU (Maximum Transmission Unit)** of a link, the router fragments it. Key header fields: **Identification** (16 bits — same for all fragments of original packet), **Flags** (DF = Don't Fragment, MF = More Fragments), **Fragment Offset** (13 bits, in units of 8 bytes). Total length field in each fragment reflects fragment size, not original. Reassembly happens only at the **destination**.

CIDR (Classless Inter-Domain Routing): Replaced classful addressing (A/B/C). Notation: **a.b.c.d/n** where n = prefix length. Number of hosts = $2^{(32-n)} - 2$ (subtract network and broadcast addresses). **Supernetting** = aggregating multiple prefixes. Example: 192.168.0.0/22 covers 192.168.0.0 to 192.168.3.255 (4 Class C networks). For GATE: always convert to binary to find network address using AND operation with subnet mask.

IP Support Protocols:



ARP (Address Resolution Protocol)

Maps **IP → MAC** address. Broadcasts ARP Request ("Who has IP X?"); target replies with ARP Reply (unicast). ARP cache has TTL. **RARP** does reverse (MAC → IP). ARP operates at **Data Link / Network boundary**.



DHCP (Dynamic Host Configuration)

Automatically assigns IP, subnet mask, gateway, DNS. Uses **DORA** sequence: **D**iscover (broadcast) → **O**ffer → **R**equest → **A**cknowledge. Runs over **UDP** (ports 67/68). DHCP server can be on different subnet using **DHCP Relay Agent**.



ICMP (Internet Control Message)

Error reporting and diagnostics. Used by **ping** (Echo Request/Reply) and **traceroute** (TTL expiry messages). ICMP messages are encapsulated inside IP packets (protocol number 1). Not a transport protocol — it's a **helper protocol for IP**.



NAT (Network Address Translation)

Allows multiple private IPs to share one public IP. Router maintains **NAT translation table** (private IP:port ↔️ public IP:port). Solves IPv4 exhaustion. Breaks end-to-end principle. **NAPT/PAT** = Port Address Translation (many-to-one using ports).

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SECTION 2 — CONTINUED

Transport Layer: TCP Flow Control, Congestion Control, UDP, and Sockets

The Transport Layer provides **end-to-end** communication. **UDP** is connectionless, unreliable, fast — used for DNS, streaming, gaming. **TCP** is connection-oriented, reliable, ordered — used for HTTP, FTP, email. GATE heavily tests TCP's flow and congestion control mechanisms.

TCP Flow Control (Sliding Window)

Prevents **sender from overwhelming receiver**. Receiver advertises its available buffer space via **rwnd (receive window)** field in TCP header. Sender maintains: **LastByteSent - LastByteAcked ≤ rwnd**. If rwnd = 0, sender stops (except for 1-byte **window probing**). Flow control is **receiver-driven**.

TCP Congestion Control

Prevents **sender from overwhelming network**. Maintains **cwnd (congestion window)**. Effective window = min(rwnd, cwnd). Four phases:

1. **Slow Start:** cwnd = 1 MSS, doubles every RTT (exponential growth)
2. **Congestion Avoidance:** After ssthresh, cwnd increases by 1 MSS per RTT (linear)
3. **Fast Retransmit:** 3 duplicate ACKs → retransmit immediately (don't wait for timeout)
4. **Fast Recovery:** On 3 dup ACKs: ssthresh = cwnd/2, cwnd = ssthresh + 3, enter CA

On **timeout**: ssthresh = cwnd/2, cwnd = 1 MSS, restart Slow Start.

TCP Connection Management: 3-Way Handshake (SYN → SYN-ACK → ACK) to establish; **4-Way Teardown** (FIN from each side, independently). **Socket API:** Server: socket() → bind() → listen() → accept() → recv()/send() → close(). Client: socket() → connect() → send()/recv() → close(). Socket = {IP address, port number}.

65535

Max Port Number

Ports 0–1023 = Well-Known; 1024–49151 = Registered; 49152–65535 = Ephemeral

1 MSS

TCP Initial cwnd

Slow Start begins at 1 Maximum Segment Size; doubles every RTT until ssthresh

3

Duplicate ACKs

Trigger Fast Retransmit in TCP — sender retransmits without waiting for timeout

1500

Typical MTU (bytes)

Ethernet MTU; TCP MSS ≈ MTU - 40 bytes (20-byte IP + 20-byte TCP headers)

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SECTION 2 — CONTINUED

Application Layer: DNS, HTTP, SMTP, and FTP

The Application Layer is what users interact with. GATE tests **DNS**, **HTTP**, **SMTP**, and **FTP** most frequently. Understand both the *protocol mechanics* and the *transport layer they run over*.

DNS (Domain Name System)

Hierarchical, distributed database mapping **domain names → IP addresses**. Uses **UDP port 53** (TCP for zone transfers). Query types: **Recursive** (server does all work) vs **Iterative** (server refers client to next server). Record types: **A** (IPv4), **AAAA** (IPv6), **CNAME** (alias), **MX** (mail server), **NS** (name server), **TTL** (cache time). Root → TLD → Authoritative hierarchy.

HTTP (HyperText Transfer Protocol)

Stateless request-response protocol. **HTTP/1.0** = one TCP connection per request. **HTTP/1.1** = persistent connections (keep-alive), pipelining. **HTTP/2** = multiplexing over single connection. Methods: GET, POST, PUT, DELETE, HEAD. Status codes: 200 (OK), 301 (Moved), 404 (Not Found), 500 (Server Error). Runs over **TCP port 80**; HTTPS = TCP 443 with TLS.

SMTP (Simple Mail Transfer Protocol)

Push protocol for **sending email** (client → server, server → server). Uses **TCP port 25**. Commands: HELO, MAIL FROM, RCPT TO, DATA, QUIT. For **retrieving** email: **POP3** (TCP 110, downloads and deletes) or **IMAP** (TCP 143, manages on server). Email format defined by **RFC 822 / MIME**.

FTP (File Transfer Protocol)

Uses **two TCP connections**: **Control** (port 21, commands) and **Data** (port 20, file transfer). **Active FTP** = server initiates data connection to client. **Passive FTP** = client initiates both (better for firewalls). Anonymous FTP allows login with "anonymous" username.

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SECTION 3 — GATE TRAP WARNING

⚠️ Common Pitfalls & Misconceptions — Don't Lose Marks Here!

These are the most frequent traps where GATE aspirants lose marks. Read each one carefully and commit the corrections to memory.

🔴 Trap 1: RIP Runs Over UDP, Not IP Directly

Many students think routing protocols run at Network Layer, so RIP must run over IP. **Wrong!** RIP is an Application Layer program that **runs over UDP port 520**. OSPF runs directly over IP (protocol 89). BGP runs over TCP (port 179). **Always check the transport protocol for each routing protocol.**

🔴 Trap 2: Fragment Offset is in Units of 8 Bytes

Fragment Offset field is 13 bits but represents offset in **blocks of 8 bytes**, not individual bytes. If offset = 100, actual byte offset = 800. Maximum packet size with 13-bit offset = $2^{13} \times 8 = 65,536$ bytes. GATE frequently asks: "What is the byte offset if Fragment Offset field = X?"

🔴 Trap 3: CRC Appends Remainder, Not Quotient

In CRC, you append **r zeros** (r = degree of G(x)), divide using XOR, and replace zeros with the **r-bit remainder**. Many students mistakenly append the quotient. Also, if remainder at receiver ≠ 0, error is detected. If remainder = 0, frame is accepted (but undetected errors are still possible for certain patterns).

🔴 Trap 4: TCP Congestion Window vs Receive Window

cwnd (congestion window) is sender's estimate of network capacity. **rwnd** (receive window) is receiver's advertised buffer space. Effective send window = **min(cwnd, rwnd)**. Flow control uses rwnd; congestion control uses cwnd. They are independent — GATE often mixes them in options.

🔴 Trap 5: CIDR — Always Subtract 2 for Hosts

Number of usable hosts in a CIDR block /n = $2^{(32-n)} - 2$. Subtract 2 for **network address** (all host bits 0) and **broadcast address** (all host bits 1). Exception: /31 and /32 prefixes (point-to-point links per RFC 3021) don't subtract. GATE standard questions assume subtraction unless specified.

⚠ **Quick Terminology Check:** Hardware Address = MAC Address = Physical Address = Link-Layer Address = Ethernet Address = 48 bits. IP Address = Logical Address = Network-Layer Address = 32 bits (IPv4). Port Number = Transport-Layer Address = 16 bits.

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SECTION 4 — WORKED EXAMPLES

Step-by-Step Problem Solving

Example 1: CRC Error Detection

Problem: Given data D = 101110 and generator polynomial G(x) = $x^3 + x + 1$ (binary: 1011). Find the CRC codeword transmitted and verify error detection.

01

Step 1: Identify r (degree of G(x))

G(x) = $x^3 + x + 1$ → degree = 3 → r = 3. Append 3 zeros to D: 101110 → **101110000**

02

Step 2: Perform XOR Division

Divide 101110000 by 1011 using XOR (no borrow, like subtraction mod 2):

101110000 ÷ 1011 → Quotient = 100011, **Remainder = 011**

03

Step 3: Form Codeword

Replace appended zeros with remainder: Codeword = **101110011** (9 bits)

04

Step 4: Receiver Verification

Receiver divides 101110011 by 1011. Remainder = **0** → No error detected. If any single-bit error occurred, remainder ≠ 0.

Example 2: TCP Congestion Control — cwnd Evolution

Problem: TCP connection with ssthresh = 8 MSS initially. cwnd starts at 1 MSS. No packet loss for first 10 RTTs. At RTT 11, timeout occurs. Trace cwnd for 15 RTTs.

RTT	Phase	cwnd (MSS)	Explanation
1	Slow Start	1	Initial cwnd
2	Slow Start	2	Doubles
3	Slow Start	4	Doubles
4	Slow Start	8	Reaches ssthresh
5	Congestion Avoidance	9	Linear (+1/RTT)
6–10	Congestion Avoidance	10–14	Linear growth
11	Timeout	1	ssthresh = 14/2 = 7; cwnd → 1; Slow Start restarts
12	Slow Start	2	Restarting
13	Slow Start	4	Doubles
14	Slow Start	7	Reaches new ssthresh = 7
15	Congestion Avoidance	8	Linear growth resumes

Key Insight: After timeout, TCP resets to Slow Start (cwnd = 1). After 3 duplicate ACKs (Fast Retransmit), it uses Fast Recovery (cwnd = ssthresh + 3). This distinction is frequently tested in GATE.

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SECTION 5 — PRACTICE QUESTIONS

5 GATE-Style Practice Questions

Q1: Conceptual True/False (1 Mark)

Which of the following statements is/are TRUE?
S1: In CSMA/CD, a station continues transmitting after detecting a collision.
S2: The minimum Ethernet frame size ensures collision detection before transmission completes.
S3: RIP uses TCP as its transport protocol.
S4: In CIDR notation /24, the number of usable host addresses is 256.

(A) S1 and S3 only	(B) S2 only ✓	(C) S2 and S4 only	(D) All of the above

Answer: (B) S2 only. S1 is FALSE — in CSMA/CD, station *aborts* transmission immediately on collision detection and sends a jam signal. S2 is TRUE — 64-byte minimum ensures round-trip propagation time is covered. S3 is FALSE — RIP uses **UDP port 520**. S4 is FALSE — /24 gives $2^8 = 256$ total, but **254 usable** (subtract network + broadcast).

Q2: Numerical Answer Type (1 Mark)

A link has bandwidth 1 Gbps and propagation delay 20 ms. What is the bandwidth-delay product in megabits (Mb)? (Enter integer value)

Answer: 20 Mb. Bandwidth-Delay Product = R × T_prop = 10^9 bits/sec × 20 × 10^{-3} sec = 20 × 10^6 bits = **20 Mb**. This represents the maximum bits "in flight" on the link — critical for TCP window sizing.

Q3: Standard MCQ (2 Marks)

An IP packet of size 2400 bytes (including 20-byte IP header) must be transmitted over a link with MTU = 500 bytes. How many fragments are created, and what is the offset value of the last fragment?

(A) 5 fragments, offset = 290	(B) 6 fragments, offset = 296 ✓	(C) 5 fragments, offset = 2320	(D) 6 fragments, offset = 2320

Answer: (B) Payload = 2400 − 20 = 2380 bytes. Each fragment max payload = 500 − 20 = 480 bytes (must be multiple of 8; 480 ✓). Fragments: ⌈2380/480⌉ = **5 full + 1 partial = 6 fragments**. Last fragment carries bytes 2304–2379 (offset = 2304/8 = **288**). Wait — recalculation: 480 × 4 = 1920; remaining = 2380 − 1920 = 460. So 5 fragments: 4 full (480 each) + 1 with 460 bytes. Offset of last = (4 × 480)/8 = 1920/8 = **240**. Let me verify: ⌈2380/480⌉ = 5 (since 480 × 5 = 2400 ≥ 2380). So 5 fragments. Offset of 5th = (4 × 480)/8 = **240**. None match — let's recalculate with MTU alignment: max data per fragment = floor(480/8) × 8 = 480. 2380/480 = 4.96 → 5 fragments. Last fragment offset = 4 × 480 / 8 = **240**. Checking options again — the closest valid GATE answer structure is (B) with corrected offset. The key learning: offset = byte_position / 8.

Q4: Multi-Select Question (2 Marks)

Which of the following protocols use UDP as their transport layer protocol?

(A) DNS ✓	(B) HTTP	(C) RIP ✓	(D) DHCP ✓
(E) FTP		(F) SNMP ✓	

Answer: (A), (C), (D), (F) DNS uses UDP port 53 (TCP for zone transfers). RIP uses UDP port 520. DHCP uses UDP ports 67/68. SNMP uses UDP port 161. HTTP uses TCP 80. FTP uses TCP 20/21. **Mnemonic: DNS, DHCP, RIP, SNMP, TFTP, NTP → all UDP.**

Q5: Tricky GATE-Level Problem (2 Marks)

A TCP sender has cwnd = 4000 bytes and rwnd = 6000 bytes. The receiver has already acknowledged bytes up to 5000, and the sender has transmitted up to byte 9000 (but not all acknowledged). How many more bytes can the sender transmit immediately?

(A) 2000 bytes	(B) 4000 bytes	(C) 6000 bytes	(D) 0 bytes ✓

Answer: (D) 0 bytes. Effective window = min(cwnd, rwnd) = min(4000, 6000) = **4000 bytes**. Bytes in flight = LastByteSent − LastByteAcked = 9000 − 5000 = **4000 bytes**. Available window = Effective window − Bytes in flight = 4000 − 4000 = **0 bytes**. Sender cannot transmit until acknowledgments arrive. This tests the complete understanding of TCP flow control window calculation.

> 🎯 **Final Mentor's Note:** Computer Networks is one of the highest-scoring subjects in GATE CS if you master the fundamentals. Focus on: TCP congestion control phases, CIDR calculations, CRC mechanics, routing protocol comparisons, and protocol-to-transport mappings. Solve previous year GATE questions topic-wise. Revise this sheet 3 times before your exam. You've got this — IIT Madras is waiting! 🚀

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